

AriMarket: Price Prediction Application of Selected Basic Commodities in Metro Manila Public Markets using the ARIMA Model

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ABSTRACT

Consumers and retailers must carefully plan their purchasing and selling strategies to navigate the ever-changing landscape of commodity prices. This unpredictability renders these strategies very important tools for staying competitive and adequately managing budgets, thus explains the need for a research study on the development of an application for a web-based commodity price predictor that endeavors to give weekly price expectations for basic commodities sold on the public markets of Metro Manila. The discussion covers the significance of the study in terms of consumer impact, business impact, entrepreneurs' impact, and the impact on the government. The use of predictive analytics, linear regression models, and the Auto-Regressive Integrated Moving Average (ARIMA) model is adopted to precisely predict the prices of basic commodities. Further, the paper emphasizes that the adjustment of model parameters improves commodity price forecasting.

This study focuses on designing a web application using the ARIMA model to predict the future prices of basic commodities sold in public markets within Metro Manila. The application processes historical price data sourced from the Department of Agriculture (DA) from January 2023 to December 2024. With the use of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), the model derives the appropriate parameters for the ARIMA algorithm, ensuring that accurate price predictions are made. The ARIMA model applies differencing when necessary and then derives the parameters to be used for its AR, I, and MA components. Model accuracy is validated using performance metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

The web application is developed incrementally, allowing for the phased integration of core functionalities, such as data visualization, ARIMA-based forecasting, and interactive tools. The system is designed around the Input-Process-Output (IPO) framework, which

processes historical price data to generate predictive insights that are displayed along with dynamic visualizations. Users can interact with the platform by browsing commodity information, viewing price forecasts, providing feedback, and accessing FAQs, thus providing a comprehensive and engaging user experience.

The results show that for most of the basic commodities traded in Metro Manila's public markets, the ARIMA model is successful in the effective prediction of future prices. MAE values are relatively low for most items, confirming robust performance in forecasting the prices of these items. For example, Well-Milled Rice attained the lowest MAE of 0.388 during high price periods within the Rice group, implying precise forecasting for stable price regimes. Similarly, Regular-Milled Rice also performs well with the MAE values of 1.373 (high price) and 0.394 (low price). Other types of rice are Premium Rice and Special Rice, in which MAE values range below 2 with a slight deviation for higher priced predictions.

In the Meat category, which has performed well, its low-price predictions have realized an MAE of under 10. Pork Kasim stands out with an especially low MAE for high prices at 1.413, showing the model restricted prediction errors for that individual item. Whole Chicken is a big challenge because it has a big MSE for low prices at 587.468, which shows that for that item, the model did not capture the variability in prices. The performance of Fish group varies. Bangus and Tilapia have fairly good predictive accuracy, MAE values for high price being 11.038 and 7.138 respectively. However, Galunggong did not perform well with its MAE being 25.460 at high price, indicating the failure of the model in depicting the price pattern of this commodity.

The results of the Fruits group have mixed performance. Banana (Lakatan) had a very low MAE of 2.127 for high prices, while Mango has significantly larger error margins; with an MAE of 29.033, this indicates difficulty in future projection of commodities that are more seasonal. From the Vegetables group, Cabbage came out

with the best MAEs for the model: it obtained values of 8.485 for the high-price and 15.539 for the low price cases. However, Tomato has the highest MAE values at 75.439 for high prices, indicating huge prediction errors due to price volatility.

In the Spices category, the lowest MAE values have been found by Red Onion to be 6.589 at high prices and 2.094 at low prices, which further indicates stable pricing trends. On the other hand, Garlic experiences greater error margins at an MAE of 26.028 at high prices, reflecting the issues associated with volatile market behavior by Garlic.

These results affirm that the ARIMA model of commodities with stable price trend robustness but reveals some limitations of the model to describe the volatility and irregular changes in some commodities. The work ahead is to introduce some extra variables or new model techniques for better predictability in the case of highly volatile commodities.

Keywords

Commodity price prediction; basic commodities; predictive analytics; ARIMA; Metro Manila

1. INTRODUCTION

1.1. Project Context

A family with five members residing in the National Capital Region needs at least ₱16,000.00 per month to meet their minimum basic food and non-food needs in 2023 - ₱2,000 higher than the national average of almost ₱14,000. This difference highlights the higher cost of living in the NCR compared to other regions. During the same year, the national poverty incidence among families was recorded at 10.9 percent—equivalent to nearly three million Filipino families without enough income to meet their basic food and non-food needs.^[25]

According to the Asian Development Bank, some of the main causes of this issue are slow economic growth, weak job creation, underdeveloped agriculture, high inflation during crises, rapid population growth, persistent inequality, and vulnerability to economic crises, conflicts, natural disasters, and environmental issues.^[24] The Department of Labor and Employment (DOLE) reports that the minimum wage in Metro Manila remains unchanged in 2024, ranging from ₱608.00 to ₱645.00 per day, the same as in 2023.^[2] This stagnation in wage rates becomes a hindrance together with the economic pressures faced by families. The middle-class earners make up close to 45% of the population in the Philippines.^[1] In July 2024, the annual inflation rate in the Philippines increased to 4.4%, rising from June's three-month low of 3.7% and surpassing market expectations of 4%. Specifically, the inflation rate concerning food rose from 6.1% in June 2024 to 6.4% in

July 2024—reflecting an increase in the cost of food items.^[5]

1.2. Purpose and Description

Consumers and retailers must carefully plan their purchasing and selling strategies to navigate the ever-changing landscape of commodity prices. For consumers, this means identifying the most advantageous times to buy, whilst for retailers, it involves determining the optimal moments to sell in order to maximize profits, reduce losses, and counteract risks. The unpredictability of price fluctuations makes these strategies crucial for staying competitive and managing budgets effectively.

To aid in this process, the application serves as a valuable tool by providing near-accurate predictions on the price of a specific commodity, most specifically the increase or the decrease of its value. By leveraging these insights, both consumers and retailers can make more informed decisions, ensuring they act at the right time to either save money or increase earnings. This predictive capability helps users stay ahead of market trends, offering a significant advantage in managing their financial activities.

For this project, the application focuses on basic food commodities, specifically rice, meat, fish, fruits, and vegetables. This application utilizes daily price data sourced from the Department of Agriculture's official website, ensuring that the data used for the application is well-monitored, reliable, and adheres to the suggested retail prices (SRP) as mandated by the Department of Trade and Industry of the Philippines (DTI). The application's prediction may be influenced by certain external factors, such as inflation rate, gross domestic product (GDP), tax rates (VAT), & oil price hikes.

Predictive analytics is the best algorithmic approach for this application because it focuses on forecasting commodity prices, which is the primary goal. By analyzing historical data, predictive analytics can generate forecasts of future price trends, empowering users to make strategic decisions. A key component in this process is linear regression, which helps in modeling and understanding the relationship between different variables and price outcomes.

Moreover, correlation analysis is vital in determining how various factors—such as fuel prices, raw material costs, labor expenses, and utility charges—affect the price fluctuations of basic commodities in the market. By identifying these correlations, the application can provide more accurate and nuanced predictions, enabling users to anticipate market changes and adapt their buying or selling strategies accordingly. This comprehensive approach not only enhances the accuracy of price predictions but also equips users with deeper insights into the underlying drivers of price movements.

1.3. Statement of the Problem

This study attempts to determine the price behavior using the proposed application that accurately predicts prices of certain basic commodities sold in Metro Manila. Specifically, it seeks to answer the following questions:

1. How accurately will predictive analytics and linear regression models forecast price trends of basic commodities and assist consumers and retailers in making informed purchasing and selling decisions?
2. How can historical data from the Department of Agriculture be effectively utilized to train and validate the price prediction model?
3. How efficiently helpful does the application serve as a basis for consumers and retailers in Metro Manila in purchasing basic commodities?

1.4. Research Objectives

1.4.1. General Objectives

The project aims to develop a web-based commodity price predictor application that delivers price forecasts for basic commodities sold in the public markets of Metro Manila on a weekly basis.

1.4.2. Specific Objectives

1. To predict the future prices of rice on a weekly basis in the public markets of Metro Manila accurately through the implementation of a web application that utilizes predictive analytics and ARIMA modeling.
2. To utilize historical data from the Department of Agriculture to train and validate the price prediction model.
3. To help improve the decision-making of consumers, businesses, entrepreneurs, and the government on purchasing basic commodities such as rice, meat, fish, fruits, and vegetables.

1.5. Significance of the Study

To ensure that various stakeholders can make informed decisions in a dynamic market environment, this study aims to develop a predictive application for real-time pricing estimates. In today's rapidly changing economic landscape, fluctuations in commodity prices can significantly impact consumers, businesses, farmers, and governmental entities alike. The following benefactors will benefit from this study:

Consumers. With the application's real-time pricing estimates, customers will be able to make more educated judgments. Customers are

better able to manage resources and plan their budgets when they anticipate price variations.

Businesses and Entrepreneurs. Businesses may minimize the danger of stockouts and overstocking by optimizing their inventory levels with the aid of accurate pricing projections. The prediction application may be used by business owners to establish competitive prices and optimize earnings. Recognizing possible pricing fluctuations might help companies create risk-reduction plans that work.

Farmers. They may use the program to help them minimize losses and bargain for higher prices by using it to assist them make educated decisions about when to harvest and sell their crops. These technologies help farmers reduce market volatility risks, create strategies like crop diversification or futures contracts, and allocate resources more wisely to optimize profits by offering information into market patterns and possible price swings.

Government. As required by RA 7581, the application can aid in the government's attempts to preserve price stability. The predictor can assist in the creation of tailored policies to safeguard consumers and maintain market fairness by offering insightful information about pricing movements.

Future Researchers. Future studies can build on this work by investigating more complex prediction models, adding more variables, and improving the precision of price projections. Subsequent investigations may broaden the analysis's focus to include other parts of the Philippines or maybe the entire world, offering a more comprehensive comprehension of the dynamics of commodities prices. More extensive price research and forecasting might be made possible by expanding the program's use to a larger variety of commodities.

1.6. Scope and Limitations

This project aims to develop a commodity price prediction application specifically designed for the Metro Manila region. The application aims to forecast future prices of essential commodities like rice, meat, vegetables, fish, and fruits by analyzing historical price data and current market trends. To model and forecast price changes, this study employs predictive analytics approaches, such as statistical models like ARIMA and SARIMA that utilize linear regression and correlation analysis. These techniques assess the impact of various factors on commodity price fluctuations, including fuel

prices, raw material costs, labor expenses, and utility charges.

Despite this, the researchers recognize several inherent limitations. This project focuses solely on the chosen commodities, and market price projections are limited to Metro Manila. The Department of Agriculture's website serves as the primary data source for this study, which means that other important data sources that might have offered a more thorough analysis may have been overlooked. Additionally, notable changes to the market or unanticipated occurrences that were not factored into the model could have an impact on the prediction abilities of the program. This specifically involves market sentiments and other relevant causes of price fluctuations such as inflation, tax rates, and the law of supply and demand. While the program includes a few aspects that influence commodity prices, it may not consider all potential variables or external impacts. As a result, the precision and dependability of the price forecasts could be compromised.

2. RELATED WORK AND TERMINOLOGY

Pricing acts as a crucial economic signal that influences market dynamics by reflecting the interaction of supply and demand factors, including competition, production costs, and economic trends. In the Philippines, where commodities like rice, beef, fish, and vegetables are essential to the economy, accurate price predictions are vital for effective economic and policy planning. Local agricultural markets are affected by both domestic production and international trade, necessitating robust forecasting methods. While traditional statistical models like ARIMA and SARIMA have been employed for price predictions, advanced techniques such as neural networks, including Bi-LSTM, have shown promise in improving accuracy. The use of hybrid models that combine various forecasting approaches has gained traction, offering enhanced predictive reliability across different markets.

2.1. Price Prediction

According to the International Monetary Fund, price is defined as the sum of money that a buyer exchanges for a good or service from a supplier.^[38] It functions as a means of exchanging goods and services among consumers in the marketplace. It also serves as a gauge for how much demand there is for a service as well as how readily available it is.^[19] Based on economic literature, market arrangements and the intensity of competitive competition have a significant impact on price. As a result, when assessing the competitive effects of business mergers and outlawed activities, pricing competition is vital.^[14] These days, prices are usually given in local currencies, which can be the British pound, US dollar, or Philippine Peso, depending on the nation.^[39]

Due to its ability to indicate supply and demand and advise producers on how much to produce, price is a vital component of the economy. This technique, called the "price mechanism," depends on prices gradually shifting until supply and demand are balanced.^[26] The price at which the quantity of an item or service supplied by sellers matches the quantity required by buyers, guaranteeing market equilibrium, is known as the "market-clearing price" and is frequently mentioned by economists.^[38] Two main elements can influence price fluctuations: production costs and market forces. Market factors like competition and supply-demand imbalances are important; a shortage or a drop in the number of rivals can cause prices to rise significantly, while an oversupply or fierce rivalry might result in a decrease in prices. However, pricing is directly impacted by production costs, which include labor, raw material, manufacturing supply, distribution, and overhead expenditures. Price changes for goods and services can result in large variances.^[17]

A key component of financial research is price change, and forecasting these changes is frequently just as important as the changes themselves.^[34] Price prediction, often known as price forecasting, is the process of projecting future prices for goods, services, or commodities by examining a variety of variables, including supply, demand, seasonal patterns, features, and rival pricing. It analyzes past pricing, economic variables, and other pertinent data using algorithms. Regression analysis is used to improve accuracy by analyzing the relationship between one or more dependent variables and independent variables. The target variable is a numerical expression that is affected by several predictors. Linear regression helps determine these effects. Both descriptive and predictive analytics are used in price forecasting; predictive analytics projects future results based on historical trends. For example, historical prices can be used to predict future pricing trends. For example, one can forecast the price of maize using previous trends.^[16]

2.2. Basic Commodities: Background, Production, and Retail Price

With 30 million hectares of land, 47 percent of which is used for agriculture, the Philippines is a prime example of an agricultural nation. The nation is endowed with rich natural resources, arable land, and a hardworking farmer workforce—all of which are necessary for a healthy agriculture industry.^[3] Commodities are defined in procurement as raw or semi-processed materials required in the manufacturing of goods.^[35] These resources, which include agricultural goods, chemicals, oils, minerals, and fuels, are transferable or consumable assets that add value when used for manufacturing or direct consumption.^[28] To be specific, agricultural goods including rice, meat, fish, and vegetables are the basic

dietary staples that are usually found in public markets.^[12] However, commodities have now come to include synthetic materials, specialty metals, and alternate energy sources.

About 80% of Filipinos eat rice as a staple diet, and it makes up a significant portion of their diet. Millions of Filipino farmers rely on it as their main source of income as it is the most significant crop in the country.^[10] The Philippines, which continues to hold the eighth-most-important position in the world rice production rankings, is among the top producers and consumers of rice worldwide, along with China and India.^[11] Rice can be grown year-round in the country due to its tropical climate, however despite domestic production, the country imports a significant amount of rice, which negatively affects farmers' lives and ranks as the third most imported agricultural product.^[36] According to PSA data from 2021, 4.81 million hectares of rice were harvested in the Philippines, resulting in a total production of 19.96 million metric tons valued at ₱403.89 billion.^[30]

Over the world, the Philippines ranks as the eighth-largest producer, importer, and consumer of pork.^[22] With US\$2.0 billion in imports, the Philippines ranked seventeenth in the world in terms of meat markets in 2021. With a 0.7% market share in the same year, the Philippines was the twenty-seventh-largest fresh meat retail market worldwide. The top three categories were pork, beef and veal, and poultry.^[7] Meat consumption has increased significantly in 2023, especially pork, which was predicted to be the most consumed meat variety by that year, averaging roughly 15 kilos per person. The typical Filipino also eats 3 kg of beef or veal and 11.6 kg of chicken, in addition to pork.^[19] Around 1.53 million tons of pork and 1.74 million tons of chicken meat were consumed per person in 2022.^[23]

Fish is vital to the Philippines' economy and food security. Fishing has always been a popular past time in this island nation with a long coastline. Fish plays a significant role in Filipino culture and customs, indicating the people's strong affinity for seafood. Because of the markets for both domestic and export fish, there is a steady demand for fish. In the second quarter of 2024, the Philippines produced a total of 1.02 million metric tons of fish. The fishing industry plays a major role in the economic growth of the Philippines, since it contributes significantly to the country's GDP, compared to the 1.08 million metric tons produced in the same quarter last year, a decline of 6.2%.^[13]

Vegetable farming provides for a significant amount of the fundamental needs in the Philippines. The Philippines' vegetable sector has a lot of promise. While the area planted to vegetables is rising at a rate of 1.7%

per year, the country produces nearly 2% more veggies annually.^[32] The Philippines has an average yield of 8.26 t/ha, a total production of 5.58 metric tons (MT), and a total plant area of over 675,726 hectares scattered throughout its three main islands, according to the Philippine Statistics Authority (2019). With its industry comprising 4.1% of the country's GDP (gross domestic product), which is derived from agriculture, it accounts for a sizeable share of the economy. The nation's overall economic production is supported by the cultivation and commerce of vegetables. The vegetable industry also supports related businesses like transportation, processing, and retail in an effort to boost the GDP even further.^[31]

2.3. Related Studies

Local research has been done on price forecasting and prediction for a few basic commodities. Notably, a 2023 study carried out at PUP examined Philippine retail rice prices and palay production. Based on past rice prices in the Philippines from 2012 to 2021, this study projected a steady increase in the retail prices of three types of rice: regular-milled, well-milled, and special rice, over the following five years, from 2023 to 2027. The statistical model used in the study was called ARIMA (Autoregressive Integrated Moving Average).^[18]

Bulacan State University conducted a study that projected the monthly retail costs of three commodities: native onions, well-milled rice, and chicken eggs. This study used a variation of the ARIMA model known as SARIMA (Seasonal Autoregressive Integrated Moving Average) to forecast an upward trend in their pricing.^[24] Furthermore, a different study conducted at PUP looked at the price behaviors of ten different commodities that are sold in public markets, including five vegetable commodities (sitao, eggplant, tomato, carrot, and cabbage) and five livestock and poultry commodities (beef brisket, pork ham, pork liempo, whole chicken, and chicken egg).^[27]

A neural network model called the Bidirectional Long Short-Term Memory (Bi-LSTM) algorithm was used in a study conducted at Isabela State University, in contrast to the ARIMA model, which is widely used for price prediction in many local studies. The prices of five food commodities—tomato, beans, cabbage, sweet potatoes, and squash—were predicted using this model in the future. Six years' worth of historical pricing data were used by the study to train the Bi-LSTM model. Consequently, the model yielded dependable one-year price projections for every commodity that was chosen, providing insightful information that can assist suppliers, farmers, and policymakers in making well-informed decisions.^[29]

On a different note, UP Diliman students' study, which did not concentrate on agricultural commodities,

forecasted oil and petroleum prices in the midst of the conflict between Russia and Ukraine. Three regression models were used in this study: Random Forest Regression (RFR), Support Vector Regression (SVR), and Multiple Linear Regression (MLR).^[21] Notably, three of these local studies used the ARIMA model, but they only looked at one or three different kinds of products. On the other hand, the suggested application seeks to offer insights concerning a wider variety of goods, such as vegetables, meat, fish, and grains.

Many studies on price prediction have also been carried out in other nations, with a special emphasis on agricultural items in those nations. Comparing various price prediction algorithms, a study carried out in Peru found that the most often used models were neural networks, specifically Backpropagation Neural Network (BPNN), Radial Basis Function Network (RBFN), Artificial Neural Network (ANN), and Wavelet Neural Network (WNN). Statistical models such as the ARIMA-MSVR hybrid model, Seasonal ARIMA (SARIMA), Vector Error Correction Model (VECM), Auto-Regressive Integrated Moving Average (ARIMA), and Multiple Support Vector Regression (MSVR) came after them. Nine studies showed that neural networks performed better than other predictive models; in those circumstances, neural networks were the most successful models. Nevertheless, it is not possible to say with certainty that Neural Networks are the best model in every situation, even with their impressive performance.^[6]

Another study comprehensively reviews and examines various methodologies for agricultural product price prediction, offering insights that may aid future research in this area. It examines three main categories of forecasting techniques: intelligent prediction models, which are excellent at identifying non-linear price relationships; combined model predictions, which combine multiple forecasting techniques to increase accuracy; and traditional methods, which concentrate on linear predictions, such as regression analysis, time series forecasting, and gray models. The study emphasizes the significance of adjusting model parameters for improved predictive performance and the necessity of combining conventional and sophisticated approaches for forecasts that are more accurate and trustworthy. The study emphasizes the need to use improved forecasting techniques as agricultural markets become more dynamic. It demonstrates that combining intelligent models for long-term predictions with linear methods for short-term trends provides a comprehensive picture of market behaviors. The study highlights the necessity of ongoing improvements in predictive models and stresses the significance of adjusting to changing market complexity to improve decision-making for agricultural supply chain players.^[33]

In a study conducted in Indonesia, researchers have presented a thorough analysis of techniques for forecasting the prices of agricultural products, emphasizing Artificial Intelligence, Data Mining, and Regression as the most popular ways. Through the use of text-mining tools and a Systematic Literature Review, the study provides insights into the frequency of particular methodologies in price prediction research. In addition, it points out gaps in the literature and suggests areas for further study, particularly in light of the difficulties the current pandemic is posing. The study intends to educate academics on the best practices and stimulate additional research into these prediction approaches in agricultural markets.^[40]

Researchers in Malaysia have developed a software program that offers comprehensive data and tools for making decisions to help farmers optimize their agricultural methods. An aspect of the software that uses LSTM algorithms for accurate price forecasts and modifiable graphics is commodities forecasting. Through interactive cards and updates, users can obtain the most recent information about agriculture, official statements, and weather forecasts. The software also provides a directory to facilitate trade opportunities by bringing together farmers and suppliers. The forecasting models, which are maintained in a backend database, are continuously improved by integrating historical and real-time data. Features including price forecasts, market updates, and supplier information are supported by the user-friendly interface. The LSTM algorithm's improved performance in handling complicated datasets serves as a testament to the software's usefulness, and more improvements are in the works to add trading and more in-depth price analysis functions.^[41]

Another study from Malaysia contrasts several approaches to forecasting agricultural prices, concentrating on two models: LSTM and ARIMA. ARIMA fared best, providing the most accurate forecasts, in the early testing with tiny amounts of data. But LSTM performed far better when bigger data sets and additional variables like humidity and temperature were included. Over time, LSTM improved in accuracy and became a superior option for price prediction. Because LSTM can handle larger and more complicated data, the research finds that it is a better alternative for long-term agricultural price prediction. Future developments will concentrate on assisting farmers in trading more effectively and improving pricing analysis.^[8]

In addition to numerous studies conducted worldwide, there are already mobile applications available for download that provide users with real-time insights into commodity prices. One of these is the Commodity Price Checker which offers consumers real-time prices for a wide range of commodities, including energy, metals, cereals, and livestock, along with a thorough study of the

US commodity market. In contrast to several platforms that focus on particular commodities or locations, this app provides a comprehensive view of the US market. Users can also monitor pricing changes over a range of time periods, from seven days to six months, which provides them with important information for well-informed decision-making. Through the integration of real-time pricing with pertinent news and statistical information, the Commodity Price Checker provides users with the necessary tools to keep informed about market trends and make wise investment decisions.^[9]

2.4. Synthesis

Pricing affects market dynamics in a more general way and plays a crucial role as an economic signal. Indicators of supply and demand, such as competition, production costs, and economic trends, all have an impact on price changes. Accurately predicting price changes has become increasingly important, particularly for commodities, and methods like algorithmic models and regression analysis are used.

Commodities such as rice, beef, fish, and vegetables are vital to the Philippine economy in agricultural contexts and account for a significant share of the country's food production and consumption. The tropical climate of the nation makes farming possible all year round, however imports are necessary for some products, such as pig and rice. Research shows that both domestic output and foreign trade have a significant impact on local agricultural markets, which makes price forecasting especially important for economic and policy planning.

In the Philippines, local research on price prediction has mostly used statistical models like ARIMA and SARIMA to anticipate the prices of agricultural goods like rice and vegetables. For more precise commodity price forecasting, more sophisticated techniques have also been investigated, such as neural network models like Bi-LSTM. By using these techniques, scholars and decision-makers can more accurately predict market trends and make better-informed choices.

Studies on price prediction have expanded the use of hybrid models and neural networks globally to more accurately forecast the pricing of agricultural commodities. Comparative analyses indicate that classic time-series models such as ARIMA are still valuable in certain situations, but neural networks do better at capturing non-linear correlations in pricing data. It is also becoming more widely acknowledged that combining these models through hybrid techniques can enhance prediction dependability in a variety of markets.

3. METHODOLOGY

Developing a web application that utilizes the Auto-Regressive Integrated Moving Average (ARIMA) model to forecast future prices of basic commodities in

public markets of Metro Manila is the general purpose of the study. This chapter emphasizes exploring the core concepts of the study, selecting the appropriate algorithms for implementation, and mapping out the detailed process for constructing the ARIMarket web application.

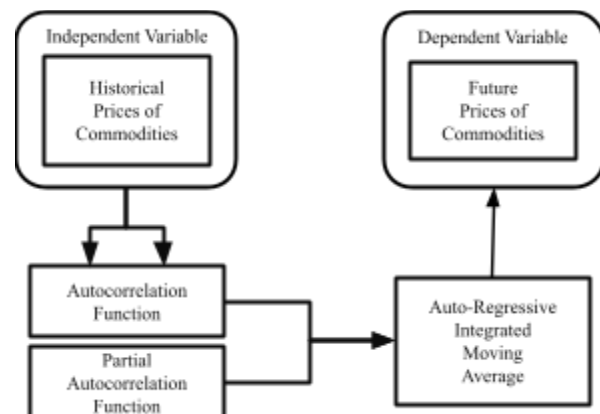
3.1. Research Design

The approach taken as the research design for this study is experimental as it relates with predictive analytics which is concerned with the analysis of historical information with the aim of anticipating future possibilities. The objective of this research is to forecast the future price trends of basic commodities through the use of the Auto-Regressive Integrated Moving Average model. In this way, the research as such seeks to test and document the correlation between the historical prices of a product and the estimated prices using the forecasting model, thus structuring the evaluation of the model's predictive performance. The experimental framework guarantees the integrity of the study since it seeks to confirm and assess the effectiveness of the model (ARIMA) in the specified conditions.

Aside from its experimental foundation, a quantitative design is used as part of the system testing. The study makes sure that the performance and dependability of the system could be assessed using numerical measures of performance such as MAE, MSE, and RMSE which evaluate the ability of ARIMA to predict objectively. It presents metrics that measure the extent of errors between actual values and predicted ones. Small values from these performance metrics indicate the model being accurate and reliable, thus higher values imply where corrections might be needed. The assessment given to the system in these cases is a strict quantitative measure, thereby giving proof that any findings from this study will hold water.

3.2. Conceptual Framework

Figure 1. Conceptual Framework



The figure above shows relationships among the variables used during the study. The independent variable will be the historical prices of basic commodities in public markets throughout Metro Manila, which are acquired from the official website of the Department of Agriculture (DA) for the period January 2023 to November 2024. These historical prices are analyzed through the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), generating visual graphs to identify the optimal parameters required for the Auto-Regressive Integrated Moving Average (ARIMA) model. Having chosen appropriate parameters, historical prices feed into the ARIMA model, which then processes and translates it into reliable forecasts regarding commodity price futures. These come to be the dependent variables used within the study.

3.3. Algorithm and Its Rules

Auto-Regressive Integrated Moving Average (ARIMA) Model

The system will use an Auto-Regressive Integrated Moving Average Model or the ARIMA Model to predict the historical prices of basic commodities for forecasting basic commodity prices. The model then follows a structured algorithm where it analyses and makes predictions of time-series data. It starts with checking on whether the data is stationary or not-this means whether the statistical properties of the data are persistent over time or not. Since ARIMA models expect to be stationary, differencing should occur to stabilize the means on non-stationary data. This is how much the data has to be differenced in order for stationarity to occur. To accomplish first-order differencing, an algorithm subtracts each observation from the last observation. If that doesn't seem to stabilize the series a little bit, a second differencing may occur.

Once the series is stationary, plot the ACF and PACF. The plots give immediate insight into the value for p-the order of an auto-regressive component-; and q-the order for a moving average component-if there are any serial dependency among observations in the time series. With these insights, the parameters are then further tuned, often using either a grid search method to explore different combinations of p, d, and q values or by applying optimization techniques to find values that best fit the model based on statistical criteria like the Mean Absolute Error (MAE) or Mean Squared Error (MSE).

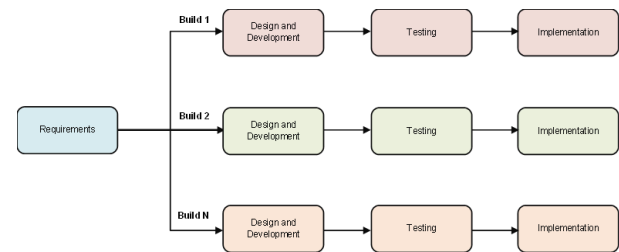
Once parameters are set, the ARIMA model merges three components: Auto-Regressive, Integrated, and Moving Average. Auto-regressive (AR) part models current value as a linear function of previous values, which have coefficients measuring the impact of each past observation. The integrated (I) part applies differencing to make the series stationery if needed. Finally, the MA part models the current value based on

past errors (or random shocks), which provides a way to account for unpredictable changes. These components together form an equation that predicts future values by adjusting both for observed patterns and random variations.

After fitting the model, it is also important to check residuals, which are the differences between observed and predicted values, to confirm that they look like white noise, which would confirm that the model has caught all the patterns in the data. The mentioned metrics from earlier, MSE and RMSE, are used to judge how well the model fits the data. Once validated, the model can then be used for forecasting, applying the ARIMA equation iteratively to make future predictions.

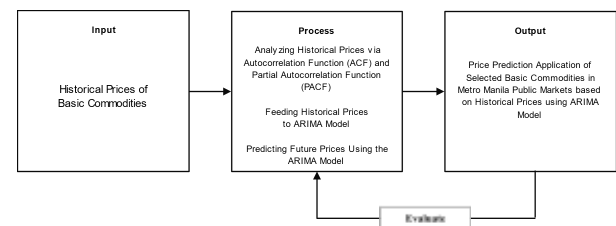
3.4. Operational Framework

Figure 2. Incremental Model Development



The development of the price prediction web application follows an incremental approach, ensuring that the system is built incrementally. This methodology is very suitable for projects that require a structured breakdown of requirements and phased integration of features. For this study, the incremental model was chosen because it is compatible with the step-by-step enhancement of functionalities. Each phase introduces a specific module, such as commodity data display, ARIMA model integration, and visualization tools, contributing to the system's overall functionality. This approach not only supports flexibility in addressing changes but also allows for continuous feedback, ultimately leading to a comprehensive and refined final application.

Figure 3. IPO Diagram

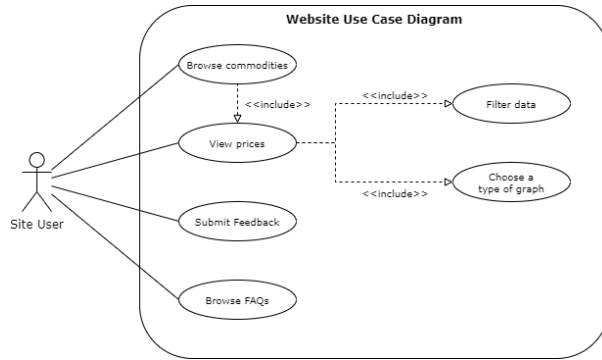


As seen in Figure 3, an IPO (Input-Process-Output) diagrammatic representation shows the workflow of the study. At the Input stage, basic commodities' historical prices are considered the input. These are processed to deduce the optimum parameters to feed into the ARIMA

model using ACF and PACF. At the Process stage, the ARIMA model is fed the data on historical prices together with the deduced optimum parameters from ACF and PACF. The Output includes the predicted prices for each commodity that the trained ARIMA model generates. This structured approach ensures that the data flow within the system is very clear.

3.5. Use Case

Figure 4. Use Case



The illustration above, Figure 4, shows in detail all the interactions that could happen between the "Site User" and all of the functionalities that are available on the website. Here, the major activities are enclosed in a circle so that a well-rounded understanding of the main operations carried out in the system may be obtained. These consist of browsing commodities, reading prices, giving feedback and seeking FAQs. Each use case represents a function that adds value to the user experience and further enables the core objective of the system-to provide accurate and comprehensive price prediction data.

One key feature is the ability for users to browse commodities. This means that the site has the capability to present users with what is on the website. It supports filtering data, where a user can narrow down search results based on certain criteria. The "include" relationship between the Browse Commodities and Filter Data use cases means that filtering is an optional but integral part of the browsing process. Further, in viewing prices, users can be allowed to choose the type of graph, whether line, bar, scatter, or candlestick charts, in which to view price data. This is represented by another "include" relationship between View Prices and Choose a Type of Graph, thus showing how the system can be able to offer different options for visualization to suit user preferences.

The diagram also provides for user feedback through the Submit Feedback use case, where users can provide their suggestions or concerns to the system. This provides a two-way interaction between the user and the website, ensuring continuous improvement. The Browse FAQs

use case also addresses common inquiries, where users can quickly access relevant information regarding the functionality, purpose, and features of the website.

The design of the system allows each use case to exist independently, giving users a choice of how they want to interact with the features they need but not necessarily in a specific order. Optional features such as filtering and graph selection add more sophistication to the user experience by bringing out more information and making the application more personalized. While the core actions and relationships are emphasized in the diagram, supplemental documentation would describe in detail the filtering mechanisms and graph selection at work.

3.6. Activity Diagram

Figure 5. ARIMarket Flowchart

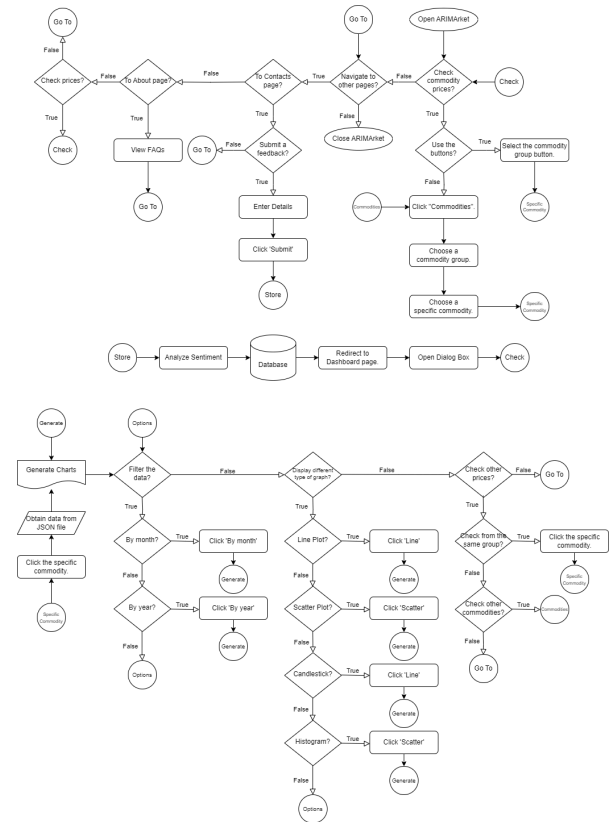


Figure 5 outlines the flow of functions within the price prediction system. The process begins with the user selecting a specific commodity through the web application interface. This action triggers the system to retrieve the historical price data for the selected commodity from a JSON file. The retrieved data is then processed using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to determine the appropriate parameters for the ARIMA model. With these parameters, the ARIMA model is trained on

historical data to generate accurate price predictions. The system subsequently displays the predicted prices, including additional visualizations such as line charts, scatter plots, histograms, and candlestick plots, providing users with comprehensive insights into price trends. This workflow ensures a seamless and interactive user experience while delivering precise and actionable information.

As discussed in the previous section, there are a few options available if the user does not need to check the price. This includes checking the About page to browse through the frequently asked questions (FAQs) or submitting feedback regarding the website through the Contacts page. This process involves the use of the database of the system, letting the user input their name and e-mail address, if necessary. The user will then be redirected to the Home page where a dialog box will appear. Its contents vary depending on the sentiment analyzed by the system if the feedback was positive or negative.

3.7. Instruments and Materials

The historical price data served as the primary input for this study, which were sourced directly from the official website of the Department of Agriculture. This dataset had detailed time-series records of the prices of basic commodities in Metro Manila public markets from January 2023 to November 2024. The use of credible and official data ensured the reliability and validity of the input, which is important for accurate forecasting. To process and analyze this data effectively, the study used data analysis software, specifically Python and its robust libraries. Some libraries in Python, such as pandas for data manipulation, numpy for numerical operations, and matplotlib and seaborn for data visualization, played an important role in transforming the raw historical data into meaningful insights. Additionally, it employed the statsmodels library in order to generate ACF and PACF plots. These graphical aids were used to determine the best parameters of the ARIMA model, which are the values of p , d , and q for auto-regression, differencing, and moving average components, respectively.

The core of this study was the implementation of the Auto-Regressive Integrated Moving Average (ARIMA) model, which served as the primary analytical tool for forecasting. The ARIMA model used historical price data and the identified parameters to predict future price trends for basic commodities. Its ability to analyze and model time-series data systematically made it a good choice for predictive analytics. Performance measurement metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) had been incorporated by the research to ensure that the actual predictions made by the model were accurate and reliable. These metrics give quantitative scores of how good the actual forecast was done by checking the actual

price values during that period. MAE measured the average magnitude of the errors in the predictions, while RMSE captured both the magnitude and variability of these errors, thus ensuring a more complete assessment of the model's accuracy.

Tools such as Python's matplotlib library and interactive charting libraries like Chart.js or Plotly were used for visualization of trends and forecasted outcomes. These tools were able to create clear and engaging visual representations, including line charts, bar graphs, scatter plots, and candlestick charts, through which users could interpret and analyze price trends. Besides the forecasting elements, the construction of the web application incorporated the web development tools and frameworks into the implementation process. Programming languages used included HTML, CSS, and JavaScript, which were used for designing the front end to provide user-friendly interactive experience. Back ends were realized using frameworks like Flask to handle server operations that involve processes such as data processing, implementing the ARIMA model. Database management systems, specifically MySQL using XAMPP, were used to store the user feedbacks, while JSON was used to store historical price data, ensuring that the system could handle large datasets while maintaining optimal performance.

3.8. Data Collection Procedures

To begin, historical price data was obtained from the official website of the Department of Agriculture (DA), particularly from the Price Monitoring Page, which is updated regularly with useful pricing information for different basic commodities. This webpage serves as a primary source for consumers, businesses, and policymakers to monitor price trends. The updates on this page are in the form of downloadable PDF documents that include both daily and weekly price data, so they provide a comprehensive view of price fluctuations across different periods.

For the purpose of this study, the researchers considered a specific subset of data, the Daily Retail Price Range. The lowest and highest recorded prices for each commodity on any given day are listed here in the Price Monitoring Page. This section is highly relevant to the analysis of price dynamics and trends. The researchers went through all available PDF documents, paying great attention to extracting the price details on each commodity. This is quite a time-consuming exercise that was deemed necessary; for only accurate data will suffice. All the price ranges-including both the lowest and the highest-were put on record with great care, and each of them has a separate CSV file as intended for each commodity. The researchers followed a step-by-step procedure to reduce possible inconsistencies in the input data entry process and thus avoid probable mistakes or omissions.

The period of data collection ranged from January 2023 to November 2024. This time period covered the most recent and up-to-date records available, giving the researchers an extended period in which to gather a robust dataset that would allow the ARIMA model enough data points to analyze trends and make predictions on future price movements. This ensured that the dataset obtained was both comprehensive and reflective of the current state of commodity pricing, thereby providing a strong basis for the development of the predictive model.

After the data collection phase was completed, the researchers moved on to the next step of the process, which involved preparing the data for use in the ARIMA model. The CSV files containing the commodity price data were used to train and evaluate the ARIMA model. The model was then trained on this data to identify underlying patterns and trends in commodity pricing, which could later be used for forecasting future price movements. To facilitate efficient integration of the data into the web application, the CSV files were subsequently converted into JSON format. This conversion was crucial, as it allowed for faster data retrieval and smoother performance within the application. The data being accessed by the JavaScript code in the web application was accessed quite seamlessly by using JSON.

Throughout the data collection and processing procedures, the researchers ensured to maintain ethical standards. Data used in this study were publicly available and obtained solely from an official and credible government website, the Department of Agriculture's Price Monitoring Page. Since the data involved no personal or sensitive information, issues on confidentiality and informed consent were not relevant. However, the researchers respected the ethics of using data in a responsible manner by ensuring that the data was handled with regard to ethical considerations and was protected from being used in ways that would violate the integrity of the study. Therefore, following such strict procedures enabled the researchers to ensure reliability, accuracy, and ethical integrity of data used in the study for its success and credibility in the predictive model.

3.9. Data Analysis

A structured approach of analyzing data for this study was carried out in order to verify the precision and performance of the proposed predictive models..

Figure 6. Mean Absolute Error: Formula

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - x|$$

Where:

n = total number of price predictions made by the system

x_i = predicted price for the i-th observation

x = actual price for the i-th observation

The used performance metric is Mean Absolute Error, or MAE, measuring the average deviation between the predictions and real values. It is often used in cases with scale-dependent data because the error on the same scale is calculated against the observed values. This characteristic makes it a good and reliable measure in judging the accuracy of the predictive models. The researchers chose MAE because it directly measures the average deviation of the forecasted prices from the actual prices. This gives a clear and understandable measurement of the model's predictiveness. By comparing the predicted values against actual values, MAE gives one very important information about the effectiveness and reliability of the model in actual practice.

Figure 7. Mean Squared Error: Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - x)^2$$

Where:

n = total number of price predictions made by the system

x_i = predicted price for the i-th observation

x = actual price for the i-th observation

Figure 8. Root Mean Squared Error: Formula

$$RMSE = \frac{1}{n} \sum_{i=1}^n \sqrt{(x_i - x)^2} \text{ or } \sqrt{MSE}$$

Where:

n = total number of price predictions made by the system

x_i = predicted price for the i-th observation

x = actual price for the i-th observation

MSE = mean squared error

Apart from MAE, other performance metrics like Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) were used to give a more general analysis. MSE computes the average of the squared differences between predicted and actual values, penalizing large errors much more severely. This is particularly useful for applications

in which large deviations have great implications, such as price prediction systems. RMSE, the square root of MSE, has the advantage of expressing the error magnitude in the same units as the original data, making it more intuitive. These measures allow for the evaluation of model consistency and give insight into the distribution of errors, especially in how large deviations affect the performance of the model. This inclusion of MSE and RMSE in addition to MAE provides a balanced evaluation that includes both the overall prediction error and the handling of significant outliers.

The analysis was done in the Python programming language, utilizing several libraries: Pandas for data manipulation, NumPy for numerical computations, and statsmodels for building and applying the ARIMA model. These tools facilitated efficient organization, analysis, and processing of large datasets. After preprocessing and structuring the collected data, the ARIMA model was applied to forecast future prices. The MAE, MSE, and RMSE were then calculated to assess the accuracy and robustness of the forecasts.

Detailed procedures were employed to ensure the reliability and validity of the analysis. Data cleansing involved careful inspection for outliers, inconsistency, and missing values in preparation for model training so that integrity of the data is upheld. To validate the correctness of the outcomes, further testing and validation operations were carried out with repetitions to limit errors and thereby improve the strength of the predictions the model made. This comprehensive approach to data analysis, underpinned by rigorous statistical methods and advanced tools, ensured the precision, reliability, and alignment of the results with the study's primary objective of accurately forecasting commodity prices.

4. RESULTS

4.1. Introduction

This chapter discusses the results of the analysis and model evaluations with a detailed discussion of the findings. Several graphs have been produced to illustrate key aspects of the data, including performance metrics and model diagnostics. These visualizations give insights into the accuracy of the ARIMA model, the distribution of residuals, and the overall effectiveness of the forecasting process. The graphs point to essential metrics of critical evaluation in Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the autocorrelation of residuals, all indicating an overall view of predictive strength of the model with specific room for improvement. Under these graphs, the basic pattern or performance characteristics are being assessed and interpreted from the ARIMA model.

4.2. Model Training Performance

Figure 9. Training Performance Comparison using Mean Absolute Error

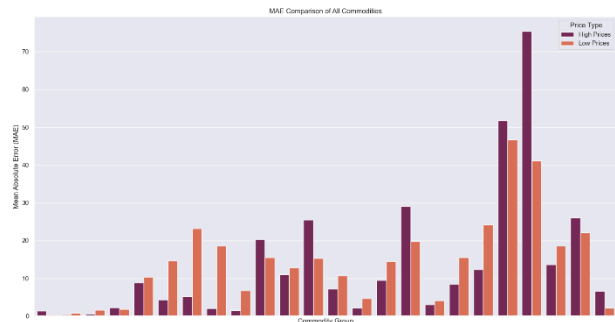


Figure 9 illustrates in detail a highly informative comparison of how MAE values vary across various commodities, bringing better clarity to predictiveness in the model performance. From the graph above, 11 specific commodities lie in MAE values below 10. This means fairly good predictive accuracy for these specific commodities. These are the commodities, starting from left to the middle: those belonging to the Rice, Meat, and Fish groups, which have error margins remarkably low, meaning that the model was able to accurately predict their values with very small deviations. This is the consistent accuracy for these items, and this means that the model is able to find stable and predictable patterns within the data. Lower MAE values indicate that the model is well-calibrated and works well for commodities with steady price trends or less volatility.

In contrast, the rest of the commodities have MAE values above 10, indicating that their prediction accuracy is lower, and the model has more difficulty in these items. Among these, three commodity groups—Vegetables, Spices, and Fruits—stand out, with certain specific commodities exceeding an MAE of 50. These commodities have particularly high error margins, indicating that the model struggles to provide accurate predictions for these items, likely due to factors such as significant price fluctuations, supply-demand imbalances, or irregular trends that are difficult to predict. The significant difference between the predicted and actual values for these commodities implies that their price dynamics are more volatile, making them more complicated for the model to predict accurately. Thus, while the model performs well for most commodities, it encounters notable challenges in predicting prices for those with greater volatility and unpredictable patterns.

Figure 10. Training Performance Comparison using Root Mean Squared Error

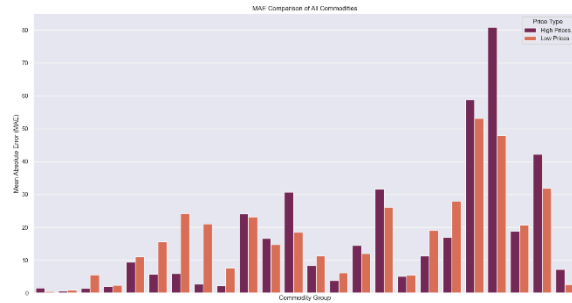
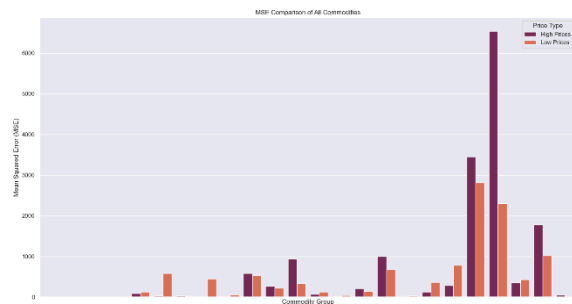
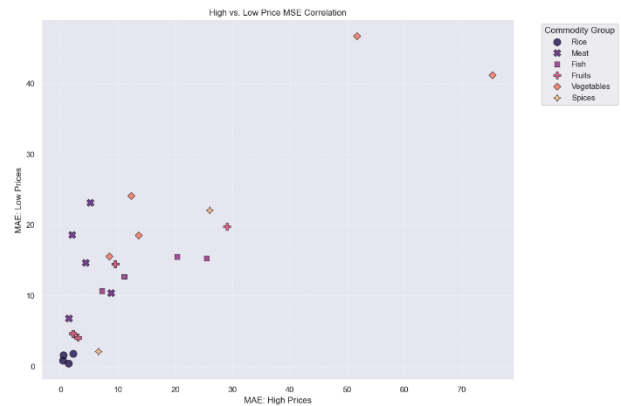


Figure 11. Training Performance Comparison using Mean Squared Error



Similarly, Figure 10 and Figure 11 showing the bar graph for the RMSE and MSE comparison respectively highlights that more stable commodities like the ones in the first four commodities which are from the Rice group, the model performs well with low RMSE and MAE values indicating predictable price trends. However, for commodities like those in the middle to the right, which are from the Fruits and Vegetable group, the model is challenged by high price volatility and outliers, which cause a larger difference between the RMSE and MAE values. These results indicate that the model has difficulty in accurately forecasting prices for more unpredictable commodities. While the ARIMA model works wonderfully for fairly stable commodities, it does not perform any better than a naive ARIMA for volatile price-changers like the ones seen in the middle-left which are meat and fish products. The highly divergent values of the RMSE and MAE for these items indicate it cannot cope well with wild moves in prices.

Figure 12. Training Performance Comparison using Mean Absolute Error



As Figure 12 shows, 8 specific commodities have MAE values below 10, implying that the model is working comparatively well on these items and that for four of those commodities—the Rice category items are substantially closer to the value of 0. This shows that for some rice types, the model will have a very small margin of error, hence providing a good prediction level. These low MAE values for these varieties of rice reflect stable and predictable patterns of data, which the model can capture well.

For all the other commodities, the MAE is greater than 10, meaning that the model failed to predict these commodities better than 10. From these, one commodity group, Vegetables, is of particular interest as within this group has two commodities with an MAE greater than 50, suggesting that it is not possible to precisely predict prices for these. These high MAE values suggest that the commodities may have price volatilities or irregular trends that are significantly large, making them difficult for the model to predict with reasonable accuracy.

Figure 13. Training Performance Comparison using Root Mean Squared Error

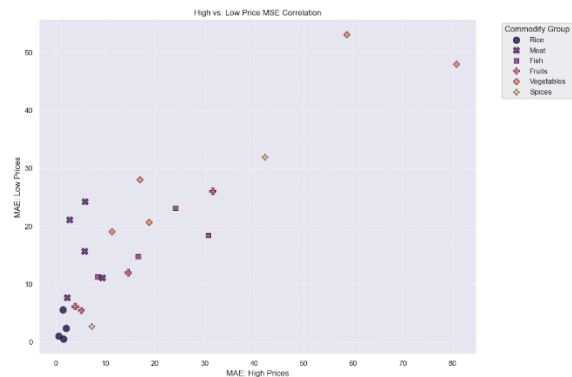
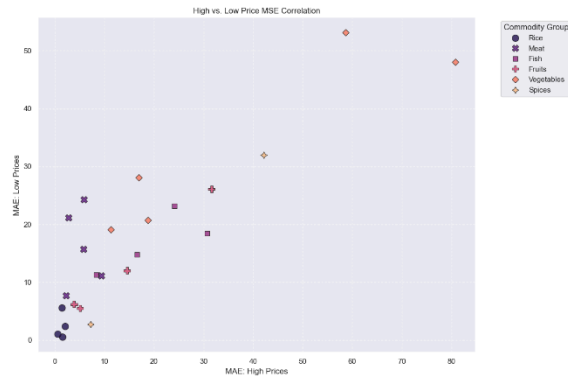


Figure 14. Training Performance Comparison using Mean Squared Error



Further results can also be seen if the MAE of the previous graph is compared against the RMSE and MSE values in Figures 13 and 14. Particularly, two of the vegetables of the Vegetables Group on the MSE and RMSE values for both high and low prices are considerably more than the MAE value. This indicates that abrupt changes in price cause heavy errors in prediction. Some commodities from the Fish Group demonstrate higher RMSE values compared to MAE, indicating that price volatility significantly affects prediction accuracy, especially for one of them, where the discrepancy is most pronounced. The same trend is observed in the Fruits Group, particularly one commodity from that group, shows a similar pattern, exhibiting large fluctuations in prices that lead to increased prediction errors, as seen in the rise of RMSE values over MAE. Finally, in the Spices Group, one of them shows significant differences between its MAE and RMSE, especially for higher prices, indicating that the volatility and outliers heavily impact the model's ability to produce accurate predictions of its prices. These results point toward the difficulties in forecasting highly volatile commodities and indicate that even more refinement of the model may be required for them to produce more accurate predictions under these conditions.

4.3. Rice Group Model Evaluation

Figure 15. Mean Absolute Error for Each Commodity in Rice Group

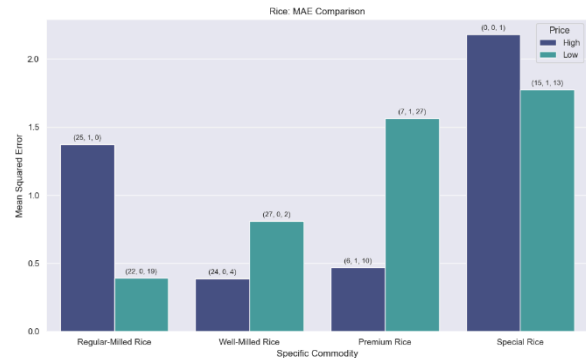
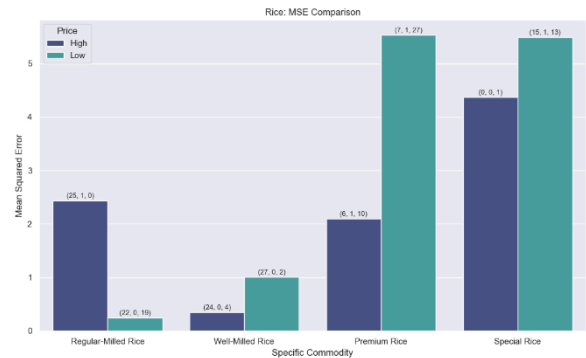


Figure 16. Mean Squared Error for Each Commodity in Rice Group



In Figure 15, the bar graph of model performance is presented for commodities falling within the Rice group, demonstrating a remarkably high accuracy value since all items that comprise this category have mean absolute error values below 2. One of them can almost just barely exceed and proves to be a clear indication that the model was perfectly capable of producing highly reliable as well as precise prediction abilities for rice-related commodities. Specifically, well-milled rice stands out with the lowest MAE value across both price ranges, particularly for high prices, where it achieves an impressively low MAE of 0.388. This low value suggests that the model performs exceptionally well in predicting the prices for this commodity, with minimal deviation between the predicted and actual values. In strong predictive accuracy, the ability of the model to have such a low error margin in predicting a high price for well-milled rice is evident.

Furthermore, other rice commodities such as "Regular-Milled Rice", "Premium Rice", and "Special Rice" also present with strong performance in the model. For all of these commodities, MAE values are kept consistently below 2 in both high and low-priced predictions, thus confirming the effectiveness of the

model in predicting price trends in rice items. MAE values of "Regular-Milled Rice" are 1.373 for high price and 0.394 for low price, and MAE values of "Premium Rice" are 0.468 and 1.563 for high and low price, respectively. "Special Rice" obtained an MAE value of 2.182 for high price and 1.775 for low price, which, even though higher, is not a bad performance.

Figure 17. Regular-Milled Rice: Predicted and Actual Prices

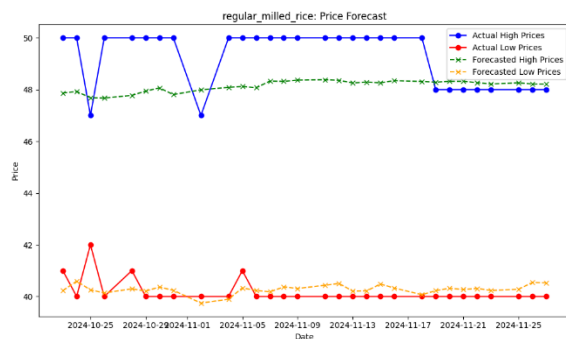


Figure 18. Well-Milled Rice: Predicted and Actual Prices

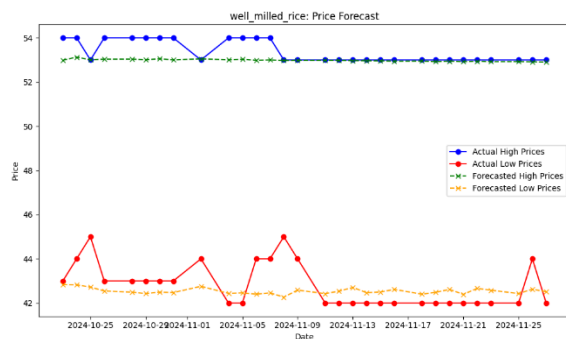


Figure 19. Special Rice: Predicted and Actual Prices

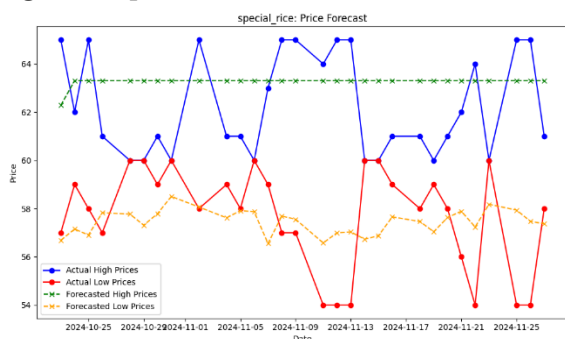
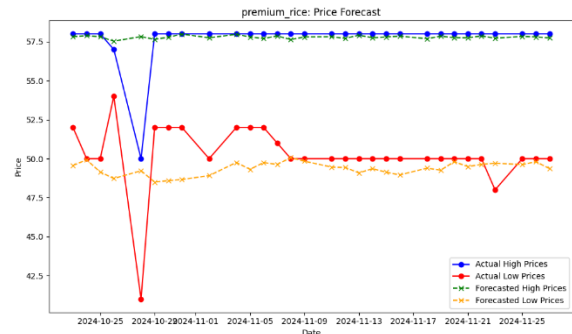


Figure 20. Premium Rice: Predicted and Actual Prices



The comparison of actual and predicted price of specific commodities within the Rice group as shown by line plots in Figures 17, 18, 19 and 20 indicates that models did a good job forecasting values close to the real figures. A major outlier however is noticed with Special Rice in the predicted prices which is shown in Figure 19. Specifically, the highest prices for Special Rice is almost constant around 63 in this period according to the predictions done by the model. However, actual prices exhibit significant fluctuations within the range of P60.00 to P65.00, introducing complex patterns that contrast with the model's learning process and affect its ability to accurately represent these variations.

This assertion highlights the fact that the model had a weakness in picking sudden sharp changes or fluctuation in price. As depicted by the figures, the models are not good at all whenever a commodity faces sharp changes which are much needed by any better representation of commodity price behavior. Therefore, this indicates the researchers need to further refine the model preferably including factors responsible for such volatility to enhance the models predictive power.

4.4. Meat Group Model Evaluation

Figure 21. Mean Absolute Error for Each Commodity in Meat Group

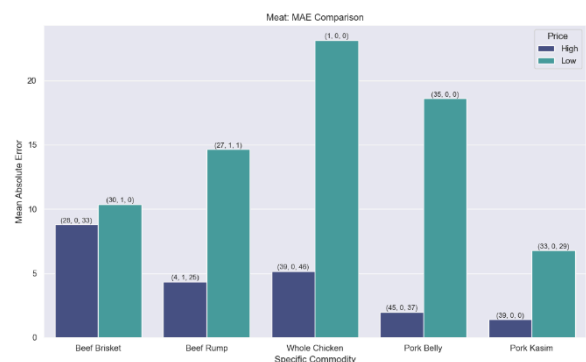
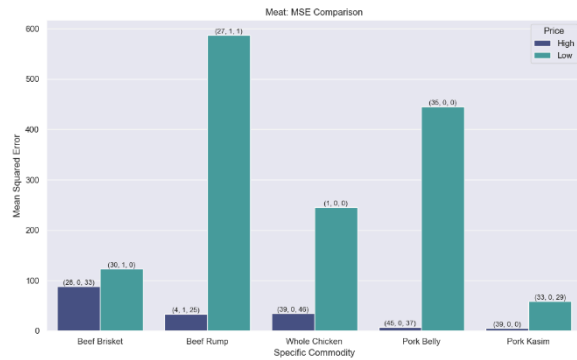


Figure 22. Mean Squared Error for Each Commodity in Meat Group



The bar graph in Figures 21 and 22 illustrates the performance of the model for the commodities within the Meat group, showing generally high predictive accuracy with most of the models at low prices having Mean Absolute Error (MAE) below 10. This gives an indication that the model is able to provide rather accurate predictions for the low prices of most meat commodities. Notably, pork kasim also stands out as the commodity with the lowest MAE value, particularly for high prices, where it achieves a very impressive MAE of 1.413. It reflects that the model excels in the prediction for the prices of this item showing minimal error both in high- and low-price forecasts in that the MAE value of the model for the high price is at 6.775. Similarly, other commodities such as the beef brisket demonstrate good performance, with MAE values of 8.785 and 10.362 for high and low prices, respectively, further reinforcing the model's effectiveness in predicting prices for these meat items.

However, one commodity, whole chicken, has a significantly higher MSE value especially at low prices where it attained 587.468. This can be interpreted as a challenge in that model since it will be struggling to predict prices of that commodity, especially in higher prices. This higher MSE indicates that the model may be having trouble capturing its price fluctuations or underlying trends, possibly due to factors such as price volatility or irregular market conditions specific to this commodity. However, its MAE value for low prices of 5.163 is relatively low, indicating that the model performs better when predicting lower price levels for this item.

Figure 23. Beef Brisket: Predicted and Actual Prices

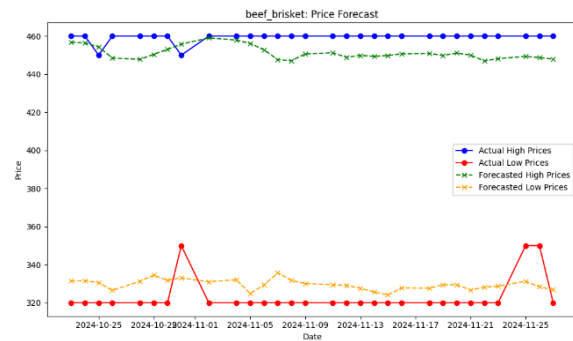


Figure 24. Beef Rump: Predicted and Actual Prices

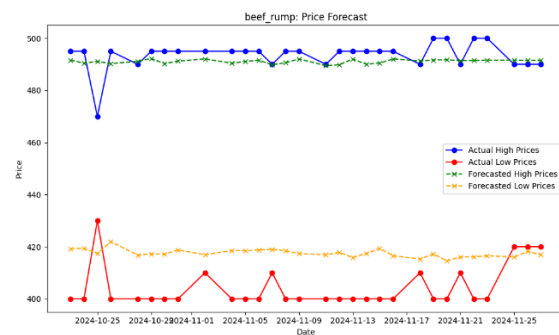


Figure 25. Whole Chicken: Predicted and Actual Prices

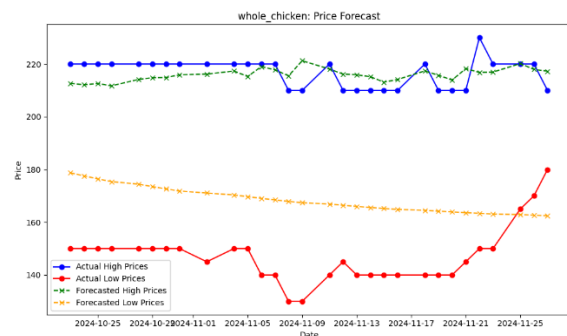


Figure 26. Pork Liempo: Predicted and Actual Prices

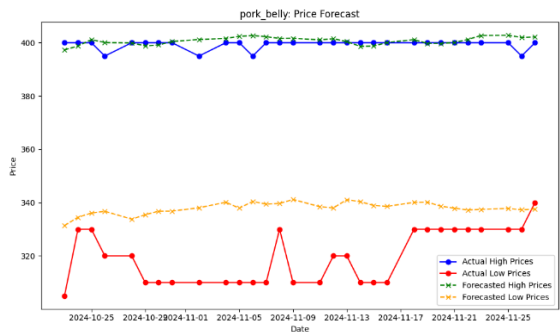
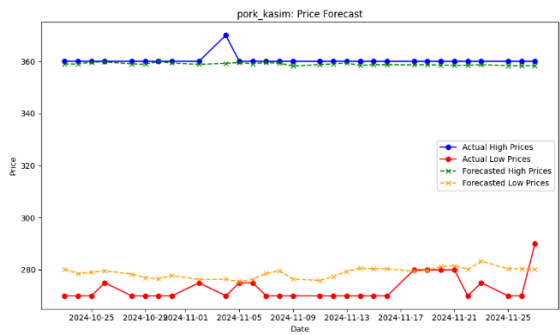


Figure 27. Pork Kasim: Predicted and Actual Prices



The high-price models for specific commodities within the Meat group do very well, as shown in Figures 23-27, with the forecasted prices hugging their corresponding actual values, indicating a good performance. It can be observed from the graph; the difference between actual and predicted values are usually around P5.00 giving confidence to the above statement. This signifies that the model can pick up the price movements of high-price scenarios effectively.

However, while the models for low-price scenarios of Beef Brisket and Pork Kasim (depicted in Figures 23 and 27) display similarly small prediction gaps, the same level of accuracy cannot be attributed to the models for Beef Rump, Whole Chicken, Pork Liempo, and Pork Belly. The models in Fig. 24, 25 and 26 seem to miss the mark a lot in providing precision as they are not able to give more concrete low-price trend predictions for the commodities they were made for in comparison to the others. This gap shows a likelihood where the features provided may have not been sufficient or they may need additional work to work well in forecasting across different price levels.

4.5. Fish Group Model Evaluation

Figure 28. Mean Absolute Error for Each Commodity in Fish Group

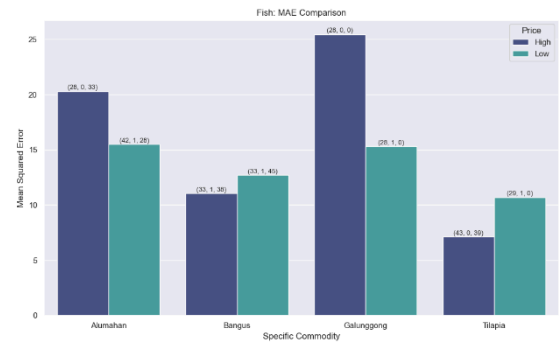
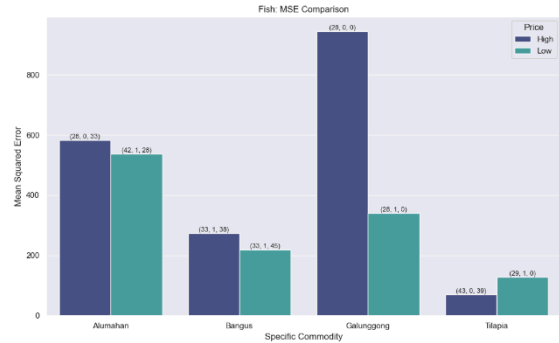


Figure 29. Mean Squared Error for Each Commodity in Fish Group



The model performance for the commodities presented in Figures 28 and 29 in the Fish group reflects a mix of strong and weaker results, demonstrating the varying levels of accuracy in predicting prices for different fish items. Some commodities, for example Bangus and Tilapia have provided excellent performance. These exhibited Mean Absolute Error values within the range of barely less than 10. For instance, while Bangus has an MAE of 11.038 for high prices and 12.702 for low prices; Tilapia has MAE of 7.138; MSE of 69.296 for high prices as well as 10.697 and MSE of 127.479 for the low prices. These low values indicate that the models for these two fish types are performing excellently, capturing price trends with high accuracy and minimal deviation. The fact that the model can predict both high and low prices for Bangus and Tilapia with such precision means that these commodities follow relatively stable price patterns that the model can easily capture.

Other than the leaders in terms of high performances, Alumahan has shown relatively good performance, with its slightly higher MAE compared with that of Bangus and Tilapia. Specifically, Alumahan has MAE at high prices amounting to 20.299, and low prices with MAE at 15.490. Although such values are somewhat higher, they remain well within reasonable performance standards

overall. An MAE value less than 20 would indicate that the model can still predict the prices for this commodity with a reasonable degree of accuracy despite the fluctuations in its price or potential volatility in the market. The higher MAE values may be due to the fluctuations in its prices, which might be caused by seasonal availability, regional disparities in price, or other factors in the market that are not predictable.

Besides that, Galunggung stands out from the other commodities within this group as illustrated by the significantly high MAE and MSE values in Figures 28 and 29. Specifically, the MAE on Galunggung is 25.460 for high prices, and the low-price MAE is at 15.312. For high-price predictions, the value is pretty high, more than any other commodity in the group – indicating that this commodity struggles a lot for an accurate price prediction. The large differences between its predicted and actual prices suggest that the model has problems capturing the complex pricing patterns associated with this commodity. There may be several reasons for this, including irregular price fluctuations, supply chain disruptions, or even seasonal demand shifts that the model fails to account for effectively. Such challenges could also mirror a need for further refinements of the model, in the integration of more factors or perhaps advanced techniques to handle price volatility.

Figure 30. Alumahan: Predicted and Actual Prices

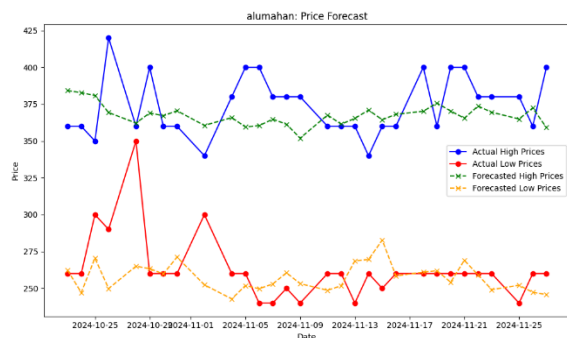


Figure 31. Bangus: Predicted and Actual Prices

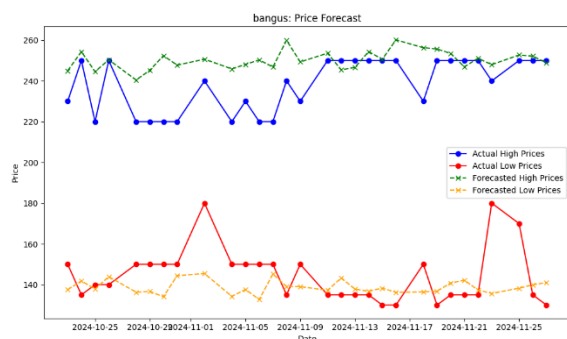


Figure 32. Tilapia: Predicted and Actual Prices

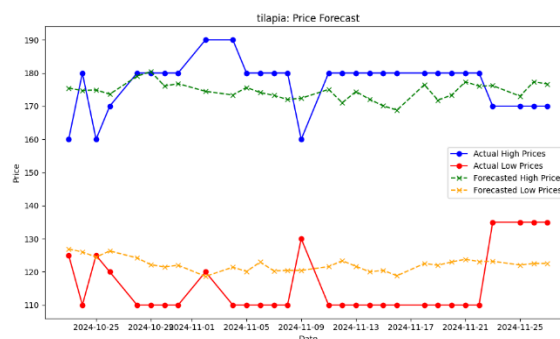
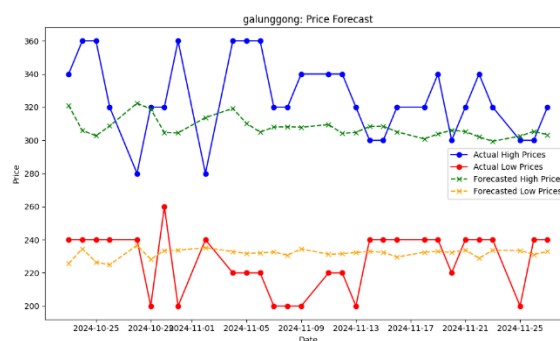


Figure 33. Galunggung: Predicted and Actual Prices



As shown in Figures 30 to 33, line graphs for the high and low price models of commodities under the Fish category: Alumahan, Bangus, Tilapia, and Galunggung show a general pattern of conformity in the way the fluctuation patterns of the actual prices and predicted prices follow each other. It means that the models catch the trend and direction of the changes in prices over time. However, despite the trend similarities, the magnitude of these oscillations is usually off the price scale. This shows an important limitation of the models as they are not even capable of mirroring the amplitude of price variations. There should be further development in predictive accuracy to boost reliability and precision.

This trend is most evident in the Alumahan and Bangus models, as depicted in Figures 30 and 31. For these products, the predicted price swings for both the high and low price ranges reflect the general upward and downward swings seen in the actual prices. These patterns indicate that the models are indeed good at identifying the overall dynamics of price changes. However, the depth and size of the fluctuation in the forecast usually vary with the actual values of the variables, which causes a mismatch in the predicted level of price. For example, while the model forecasts periods when the price of the variable is rising or falling, it does not fully depict the extent to which the price is rising or falling. This means the predictions will not closely fit the

data. This limitation underlines the need for model enhancements in order to better account for the variability and complexity of market conditions.

Conversely, however, there are a few models that demonstrate higher inaccuracies and need more improvement. For instance, the low-price model for Galunggong, as illustrated in Figure 33, produces almost a constant price prediction over time. This result is at stark contrast to the actual price data, which depicts noticeable fluctuations. This could mean that the model does not actually reflect the dynamic movement in prices of this commodity. Such inaccuracy especially seems to appear in lower ranges of prices.

4.6. Fruits Group Model Evaluation

Figure 34. Mean Absolute Error for Each Commodity in Fruits Group

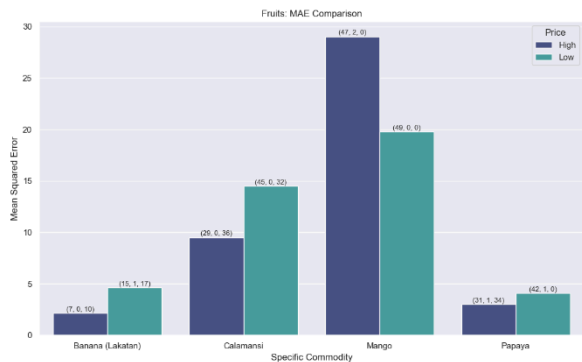
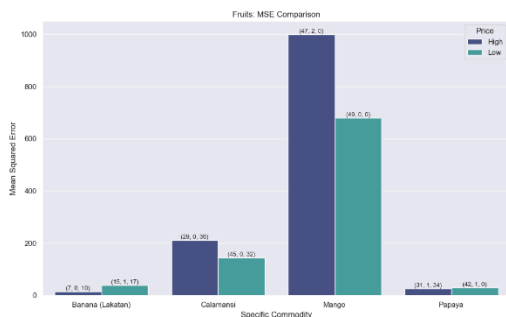


Figure 35. Mean Squared Error for Each Commodity in Fruits Group



Figures 34 and 35 report model performance for the commodities of the Fruits group, a mix of strong and weaker results, reflecting mixed levels of accuracy in price predictions. Most of the commodities of this group have Mean Absolute Error (MAE) values just under 10, suggesting good performance of the models to capture price trends for those items. Banana (Lakatan) is one of the commodities with the lowest MAE values in this group, having an MAE of 2.127 for high price and 4.633

for low price. Papaya also comes out to be the commodity that performs much better compared to the other commodities belonging to this group, showing an MAE of 2.995 and MSE of 25.794 in case of high prices. In case of low prices, its MAE is 4.077 and MSE is at 30.205. These low values suggest that the model does great in these commodities, which indicates that it is capable of giving a high level of accuracy in predicting high prices and low prices. Therefore, this means that the models for these commodities have the ability to predict prices with minimal deviation, hence it is relatively stable in prices, making it the best fit for the model.

Another commodity in the Fruits group, Calamansi, has a little higher MAE and MSE values. MAE for high prices and low prices are 9.496 and 14.499, respectively. MSE values are 211.665 and 144.201 for high and low prices, respectively. All these values indicate that for these commodities, the model still performs reasonably well but with a little more variation than for Banana and Papaya. Although this comes at higher error margins, the model still picks up price trends for Calamansi and Papaya relatively well with their MAE values staying under 15, hence having a good performance generally.

However, Mango stands out as a notable exception in the Fruits group since its MAE values are significantly higher. Its model's performance shows a big difference between the predicted and actual prices, with an MAE of 29.033 for high prices and 19.766 for low prices. The high value for the model of high prices signifies that the model is unable to precisely predict the high prices for this commodity, and a considerable error exists in the values predicted. This can be attributed to a large variation in mango prices with respect to which the model cannot capture the situation in a proper manner. This might be due to factors such as seasonal price changes, regional price disparities, or unpredictable market dynamics, hence its high MAE value. This means that perhaps the model needs to be refined further, perhaps adding additional variables or using more complex techniques to account for price volatility.

Figure 36. Banana (Lakatan): Predicted and Actual Prices

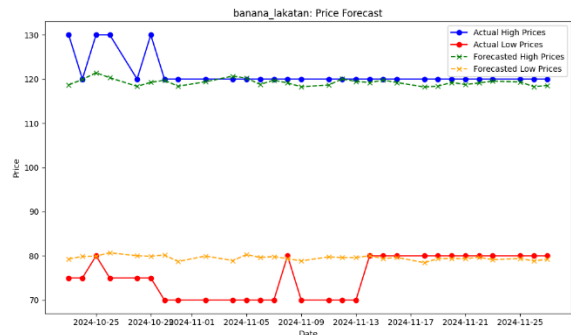


Figure 37. Calamansi: Predicted and Actual Prices

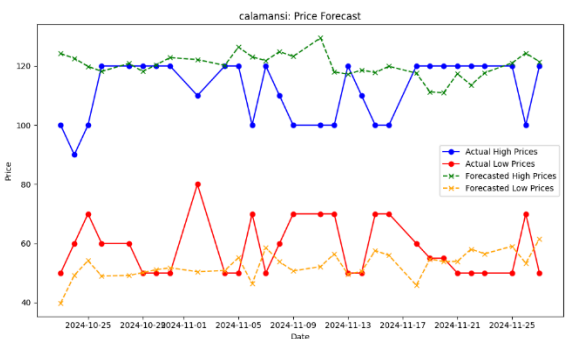


Figure 38. Papaya: Predicted and Actual Prices

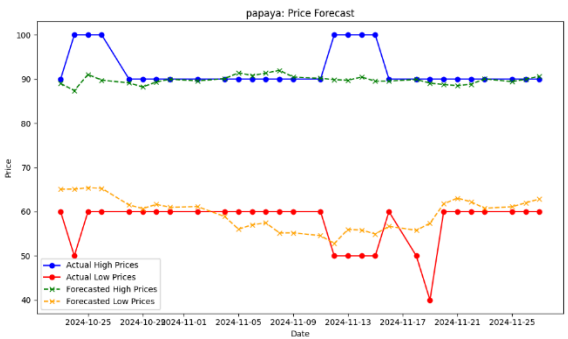
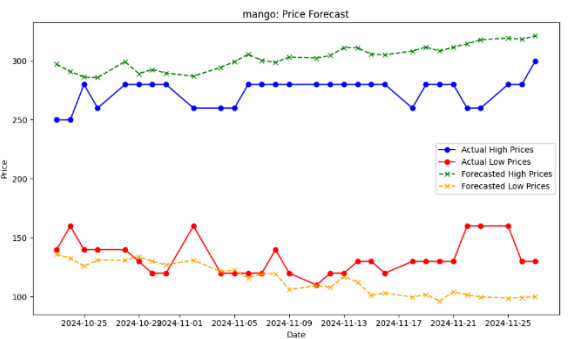


Figure 39. Mango: Predicted and Actual Prices



As shown in Figures 36 to 39, the line graphs for commodities in the Fruit category high and low price models—namely Banana (Lakatan), Calamansi, Papaya, and Mango—bring into sharp focus significant observations of the models' predictive capability and limitations. Figures 36 and 38 present the actual price data for Banana (Lakatan) and Papaya, respectively. In these, one can observe mainly stable price levels with occasional sharp jumps happening on very few specific days. These long periods of stability together with sharp changes make the models face a serious challenge during training. This is where the sudden nature of the price movement creates challenges for the models to really learn and make predictions upon the fluctuations, making their estimations not as accurate as expected.

The inability to adequately account for these sudden price shifts is a major limitation that adversely impacts the models' overall performance for these commodities. Specifically, whereas the models may approximate the general stability in the prices of Banana (Lakatan) and Papaya, they fail to predict the timing and intensity of the abrupt fluctuations accurately. That limit therefore brings it to light that probably such models have over-learned much more on the prevalent stability presented in the data so result in an underemphasis about significant and seldom encountered changes in prices.

On the other hand, the models for Calamansi and Mango have a relatively better ability to capture the timing of price fluctuations as presented in Figures 37 and 39. Though this is a positive development, it has to be noted that the MSE for these two commodities is higher than that of Banana (Lakatan) and Papaya. This is probably because Calamansi and Mango models are more responsive to changes in price trend but cannot very well estimate the magnitude of such change. Although these models successfully pick up when prices are likely to surge, they tend to miss the exact amount and, thus, the actual amount of such price variations can be way different from the predicted prices.

This is exactly what had been seen for the category of Fish, where the calculated magnitude values of price rise and falls were significantly away from those observed. This model only manages to capture the trend as well as the direction, but the loss in preciseness in giving an approximate intensity of price movements decreases the practical application value towards precise price forecasts.

4.7. Vegetables Group Model Evaluation

Figure 40. Mean Absolute Error for Each Commodity in Vegetables Group

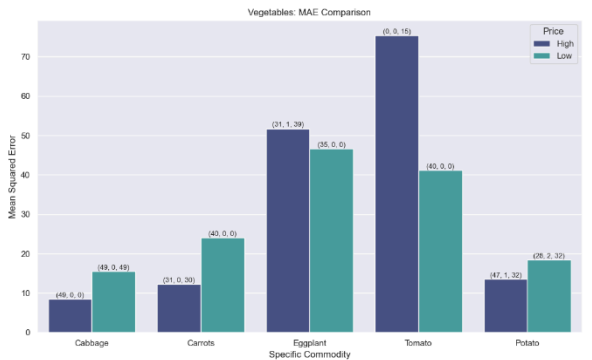
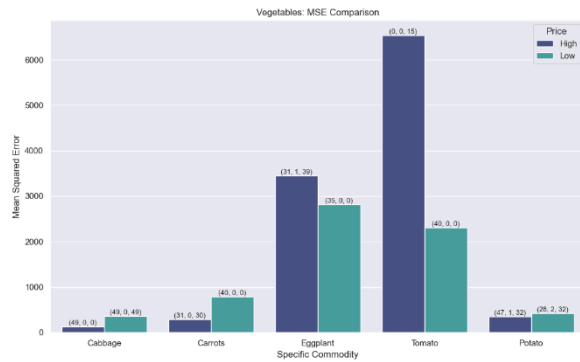


Figure 41. Mean Squared Error for Each Commodity in Vegetables Group



In Figures 40 and 41, model performance for the commodities within the Vegetables group ranges over a series of MAE values, indicating different accuracy levels across the items. All commodities in this group present an average MAE value between 10 and 50, suggesting moderate model performance in general. Within the group, Cabbage represents the lowest MAE values; thus, it presents the best model performance. Specifically, the MAE for high prices is 8.485, and for low prices, it is 15.539. Its MSE values are also relatively low, having a value of 128.492 and 362.562, respectively. These values suggest that the model performs fairly well in predicting the price trends for this commodity, with relatively low errors in both high and low-price predictions. While this performance is good, it still is a sign that there may be some fluctuations in the market for this commodity, thus causing moderate deviations in the predictions.

Other products in this category, Carrots and Potato, have higher MAE values, ranging from the high 10s to the low 20s. Potatoes have an MAE of 13.611 for high prices and 18.514 for low prices, while Carrots show an MAE of 12.319 for high prices and 24.108 for low prices. These results indicate that the model still performs moderately well for these commodities but that there are more significant deviations in its predictions compared to Cabbage. The higher error margins might imply that market conditions for these vegetables are more volatile generally, making it harder for the model to predict their price movements at the same level of accuracy.

The commodity with the largest MAE values in this category is Tomato, whose predicted and actual prices differ by a large margin. The model for Tomato has an MAE of 75.439 for high prices and 41.173 for low prices. Such a large deviation points towards a relatively poor performance in the prediction of tomato prices, especially for high prices, where the model seems to struggle the most. High MAE may be due to some big price swings or extraneous factors like supply chain interruptions, seasonal effects, or a surge in demand, which the model does not handle well. All in all, the

model works relatively well for most of the vegetables except for Tomato, for which there are significant prediction errors, and more might need to be added to account for such a commodity with precision in forecasting. These values are the highest MAE values out of all the commodities, indicating poor model performance.

Figure 42. Cabbage: Predicted and Actual Prices

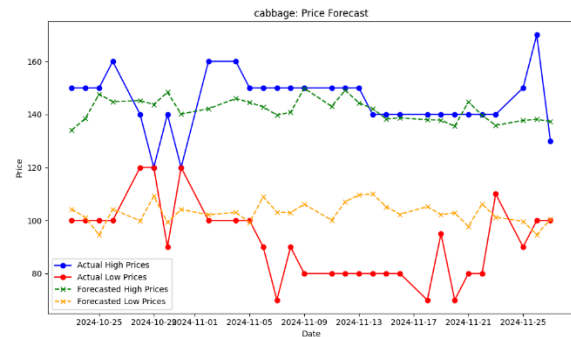


Figure 43. Carrots: Predicted and Actual Prices

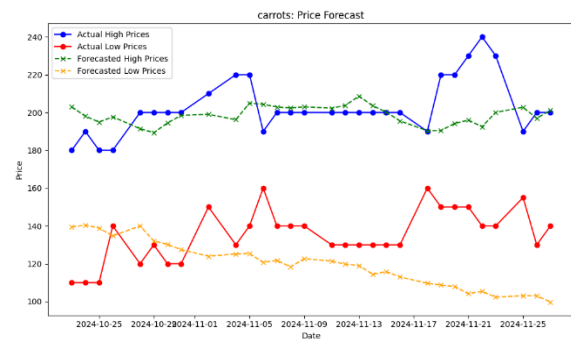
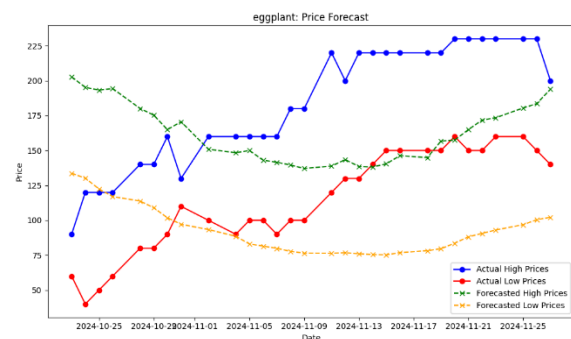


Figure 44. Eggplant: Predicted and Actual Prices



**Figure 45. White Potato:
Predicted and Actual Prices**

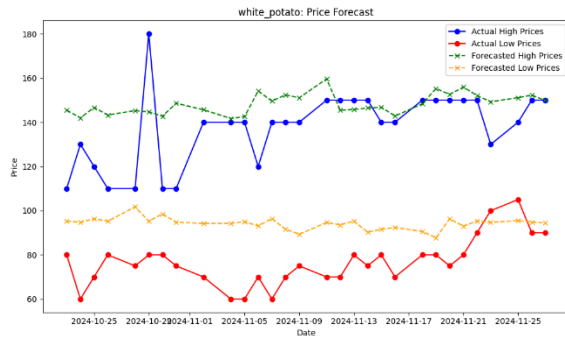
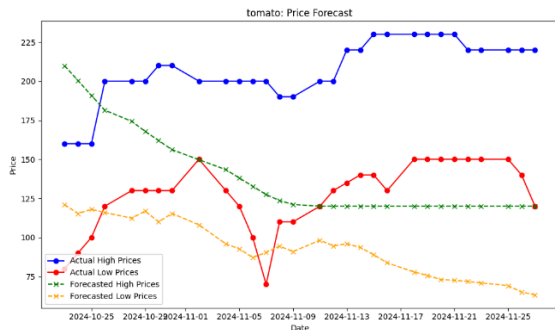


Figure 46. Tomato: Predicted and Actual Prices



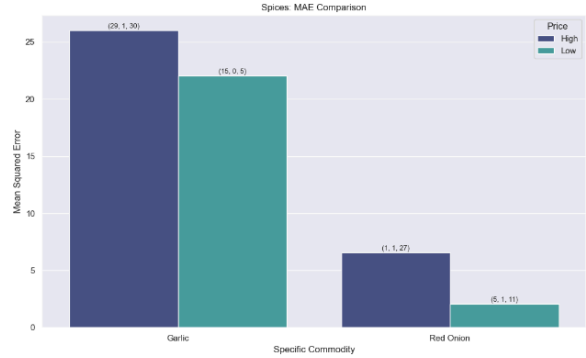
Based on the line graphs illustrated in Figures 42 to 46, the general performance of the specific commodities under the Vegetables group are mixed. Although Cabbage, Carrots, and White Potato may have better performance as seen in Figures 42, 43, and 45, the same cannot be said for the remaining commodities specifically Carrots, Eggplant, and Tomato shown in Figures 44 and 46. The predicted prices of Cabbage and White Potato have closer values to the actual prices, especially the output of the high-price models. As discussed in the previous commodity groups, the models failed to capture the magnitude of the fluctuations, making the models produce less accurate predictions yet still considered closer to the actual values than they should have been.

On the other hand, the way the predictions went for Eggplant and Tomato was not good. For the models of Eggplant, the predictions went down and up again in almost straight curve pattern at the end of December, whereas the actual prices actually went up little by little - although it managed to predict a closer value at the end of November. This issue is more serious with the models for Tomato as the curves were more drastic and way more off compared to the actual prices, especially in the high-price model. In contrast to the actual prices going up slowly, its predictions made a straight curve going

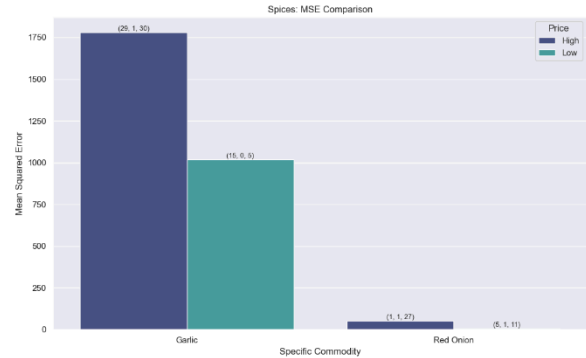
downwards first before having a constant value around P325.00.

4.8. Spices Group Model Evaluation

**Figure 47. Mean Absolute Error for Each
Commodity in Spices Group**



**Figure 48. Mean Squared Error for Each
Commodity in Spices Group**



Figures 47 and 48 show the bar graphs above depicting model performance for commodities in the Spices group, showing a mixed outcome. The lowest MAE values in this group are found for Red Onion, with a value of 6.589 for high prices and 2.094 for low prices. Its MSE values are also significantly low, with values of 52.528 and 7.166, respectively. These relatively low values indicate that the model is performing moderately well for this commodity making accurate predictions with minimal deviation from the actual prices. The model captures the price trends for Red Onion with reasonable accuracy, suggesting that the pricing dynamics for this commodity are relatively stable and predictable, allowing the model to provide a good forecast.

On the other hand, for Garlic, MAE is much higher with a high price MAE of 26.028 and low-price MAE of 22.061. The mismatch between the predicted and actual prices for "Garlic" indicates that the model does not perform well in the prediction of price trends of this commodity. The high volatility in its prices may therefore be contributing to the relatively high error margins, where the model finds it difficult to precisely predict the

price movements. This makes the model's performance for it quite notably weaker than that for Red Onion, which reflects an inherent challenge in predicting these spice price movements.

Figure 49. Garlic: Predicted and Actual Prices

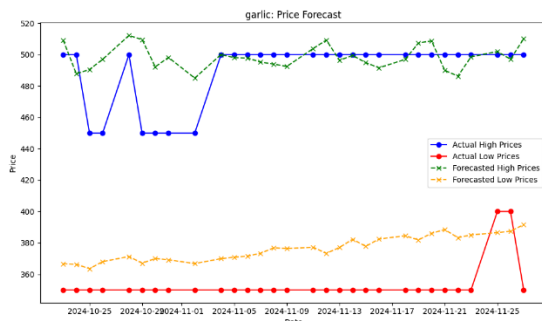
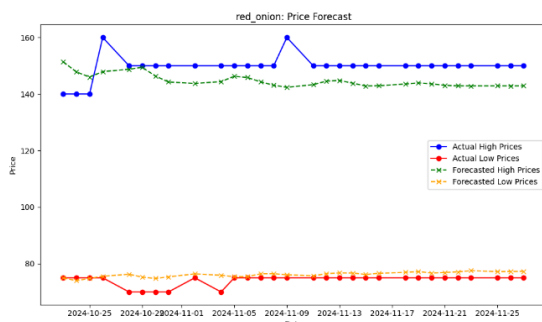


Figure 50. Red Onion: Predicted and Actual Prices



As illustrated in Figures 49 and 50, the performance of the models for the specific commodities under the Spices group—namely Garlic and Red Onion—shows mixed results. The models for both the low and high predicted prices of Red Onion show good performance with closeness to the actual values. In contrast, the models for Garlic, specifically the low-price model, show notable inaccuracies. The low-price forecasts for Garlic indicate that there are gaps of between 0 and 20 units between the predicted and actual prices, meaning there is a significant gap.

Although the model with high prices for garlic shows slight performance, especially during capturing of a general trend of trends of variations in price, they fail to bring out their scale at exact magnitudes. In that regard, it will mainly be noticed among the early days of making a forecast because the actual trend may capture the movement pattern direction while failing in reproducing an exact variation magnitude scale. This issue, typical of the issues seen in other commodity groups, is where the magnitude of price variations is not fully captured and hence the challenges in achieving accurate predictions.

The low-price model for Garlic faces extra challenges. The forecasted prices remain relatively stable over most of the forecast period, with only the last three days showing significant changes—an upward spike followed by a sharp rise of around 100 pesos. Although this behavior allows the model to approximate the highest low price recorded during the last three days, the performance is poor due to the inability of the model to reflect the dynamic price changes observed in the actual data.

4.9. Comparison with Previous Research

In a research study conducted in a university in Isabela, the Long Short-Term Memory (LSTM) model developed by Lagarteja et al. obtained a Root Mean Squared Error (RMSE) value of 2.07. Its performance was markedly better than that of the researchers' ARIMA-based model. The former had RMSE values of 80.868 and 47.994 for high and low price predictions, respectively (Lagarteja et al., 2022). This clearly demonstrates the prospect that LSTM models have towards tackling complex time-series data especially for commodity price forecasting.

Another comparable challenge was encountered by the price prediction system developed using the ARIMA model in Manila. According to the paper, students in Manila face comparable challenges in forecasting prices mainly due to the high volatility nature within their dataset. In this instance, it was characterized as sharp, unpredictable price variations, with the periods being short such that stable predictions are difficult to produce. Furthermore, the dataset for the Manila study contained price ranges with some values rounded to the nearest tens or hundreds, which made the precision of the predictions even more complicated. (Ativo, 2023) This lack of granularity posed serious obstacles for models requiring detailed numerical input to generate accurate forecasts, especially for commodities with frequent and significant price variations.

On the other hand, a model implemented in Bulacan State University utilized both ARIMA and Seasonal ARIMA, known as SARIMA, models that gained remarkable differences in performance. For example, the best SARIMA model for Well-Milled Rice in the Bulacan study was 4.231 MAE (Balilla, 2023). However, the ARIMarket system developed by the researchers demonstrated superior accuracy with significantly lower MAE values of 0.388 and 0.810 for high and low-price predictions, respectively. Similarly, for Red Onion, the best SARIMA model reported an MAE of 9.783, whereas the ARIMarket system outperformed it with MAE values of 6.589 and 2.094 for high and low-price models, respectively.

5. DISCUSSION AND CONCLUSION

This research focuses on developing a solution to address the challenge of predicting price fluctuations for

basic commodities in Metro Manila. The primary objective is to create a web-based application capable of delivering accurate weekly price forecasts to support consumers and retailers in making well-informed purchasing and selling decisions. To achieve this, the study explores the use of predictive analytics and ARIMA modeling to analyze historical price data sourced from the Department of Agriculture. Specifically, the research aims to determine the accuracy of the prediction models, assess the effective utilization of historical data for model training and validation, and evaluate the application's role in enhancing decision-making for key stakeholders, including consumers, businesses, entrepreneurs, and government entities. Through commodity products of rice, meat, fish, fruits, and vegetables, the research study endeavors to produce an effective decision-making tool, enhance the price's awareness, and help make wiser economic decisions.

5.1. Key Findings

The results of the model evaluation show that the ARIMA model is highly effective for price forecasting of certain commodities, given the steady trends in their prices. For some of these commodities such as Well-Milled Rice, Pork Ham, and Bangus, the model managed to achieve low MAE for most of the periods it was run. This performance shows that the ARIMA model is able to capture and predict stable patterns in historical price data, making it a reliable tool for forecasting in less volatile markets.

However, this model faces particular difficulties in being used for products whose price behavior is quite volatile, as seen in products from the groups of Vegetables, Spices, and Fruits. For those items, the MAE values were quite a bit larger, as if trying to price commodities with extreme short-period sharp variations. This was emphasized in the findings since volatility remains a critical factor influencing the accuracy of the forecasting models since it introduces irregular trends that are difficult to capture using traditional time-series analysis. Overall, the model shows good potential for price forecasting in stable markets but needs refinement to better address the challenges posed by highly volatile commodities, particularly in ensuring more accurate predictions for outlier-rich data.

By using MAE, MSE, and RMSE, the study provided a deeper evaluation of the strengths and weaknesses of the model about various commodity groups. Indeed, this comprehensive evaluation portrayed the ARIMA model for precise forecasting in specific applications while underlining the needs for alternative or hybrid techniques in dealing with highly volatile data. This enhances understanding of the influences behind model performance and further improvements toward increasing the reliability of the forecast under varying conditions of the market.

5.2. Interpretation of Results

The study showed particularly robust forecasting results for the Rice group, where all commodities had exceptionally low Mean Absolute Error values, indicating very accurate price predictions. Of these, Well-Milled Rice stood out with an MAE of just 0.388, showing the model's robustness in capturing stable price patterns for this staple commodity.

Similarly, the Meat group performed well with generally good predictive accuracy. Pork Ham was the top item in this category, with an MAE of 1.413, meaning it is a reliable forecast for this commodity. On the other hand, Whole Chicken had higher error margins, which means that factors that cause its price fluctuations are either more complex or less predictable, making it harder to predict for the ARIMA model.

In contrast, the Fish group performed unevenly. There were other commodities like Bangus and Tilapia for which their forecasts were well accurate having low MAE values; however, commodities like Alumahan and Galunggong exhibited higher error margins when compared to that of actual values, due to either higher price volatility or their irregular trends as handled by this model.

The Fruit group was characterized by significant variability in forecasting performance. For example, Banana (Lakatan) and Papaya performed outstandingly well, where the models were highly accurate. However, Mango exhibited considerably higher MAE values, indicating high prediction errors. This variation may indicate that there are other external factors that must be considered, such as market demand, to improve the model's predictive performance for such commodities.

The Vegetable group had a moderate forecasting success with mixed accuracy in the commodities. Cabbage showed relatively good performance with MAE values that were quite manageable. However, the error margins for commodities like Eggplant and Tomato were much higher at 51.767 and 75.439 for high prices, respectively, which indicates poor model performance. The huge differences between the predicted and actual prices indicate that it is difficult to forecast Garlic's volatile pricing trends. This volatility probably contributed to the model's inability to generate precise predictions, thus making Garlic the most difficult commodity to forecast accurately within the Spices group.

Results for the forecast of prices in the ARIMA model for the group of Spices showed a mixture of performances. Red Onion predicted relatively well, implying stability and predictability in its pricing trends. Then, it can be observed through the model that this performed reasonably well. Conversely, MAE scores for Garlic were drastically high when indicating that the

model fought to predict price movement and that the type of spice was subject to high frequency and abrupt change within the prices.

5.3. Implications of Findings

The developed system is able to accurately forecast prices for high-value commercial crops, thus offering a wide range of stakeholders benefits such as farmers, agricultural investors, and government agricultural agencies. It can help them make informed decisions about when to plant crops at the most opportune times by providing them with reliable price predictions. This insight has more value especially for farmers since it optimizes crop yield and market return, thereby minimizing the risks accompanying the fluctuation in market prices. Agricultural investors can use them to make strategic investments in the agricultural sector while government agencies utilize this data to provide an effective policy recommendation and advisory service to farming communities.

The results of this study are expected to form a basis for a much more extensive and holistic agricultural endeavor based on price forecasting of high-value commercial commodities. This larger effort will expand the scope of price forecasting to cover a wide range of HVCCs that are deemed essential to the local agricultural economy. Through integration into the forecasting model, this system will offer even better insight to the market trend, helping the stakeholders involved in the agricultural value chain.

Moreover, a typical problem which provincial agricultural offices face is the absence of a comprehensive approach to price data storage and management. Price data in most regions is not always collected consistently or is simply fragmented. This results in difficulties tracking trends and therefore making proper decisions. The results of this research are critical in addressing this gap because they will help build a robust, organized system for securely storing and managing price data. Such a system will not only improve the accessibility and accuracy of data but also enhance the efficiency of agricultural planning and policy formulation. The system will therefore contribute to a more transparent and data-driven agricultural sector by allowing better management of data, hence supporting sustainable growth and development in the industry.

5.4. Limitations of the Study

Despite its notable achievements, ARIMArket still has limitations. The current focus is limited to Metro Manila, which might limit its applicability to other regions with different market dynamics, which might limit its applicability to other regions that may have different market behaviors, pricing trends, and economic factors.

In addition to that, the system relies heavily on specific data sources, primarily from the Department of Agriculture (DA). While these sources provide valuable historical data, they may not always ensure comprehensive or real-time coverage. The dependency on a single data source limits the system's ability to adapt to sudden market changes, such as supply chain disruptions, seasonal variations, or global economic factors. Adding in other, more diverse sources of data or real-time data streams could greatly improve ARIMArket's predictive capabilities.

In addition, the ARIMA model has limitations of its own. For commodities with stable price patterns, the model is quite efficient. However, it falters when dealing with volatile commodities or irregular price variations. This limitation can be seen in the system's failures with some commodity groups like Spices and Vegetables. Future improvements may be achieved by including more complex techniques such as hybrid models or machine learning algorithms that will better capture volatility in commodities.

5.5. Recommendations

To further improve the ARIMArket system and to extend its reach, a number of strategic recommendations are suggested. First, extending the date by adding more commodities, and data to the system for each commodity would make the system more dynamic and current in its view of commodity price fluctuations. Such an integration of broader data inputs would ensure that the system captures micro and macroeconomic factors that influence commodity prices, thus making its forecasts more robust and comprehensive.

Extending the geographical reach of ARIMArket to regions outside Metro Manila constitutes another important improvement. To this end, scaling up the system to cover all other regions of the country can help ARIMArket present a national viewpoint of commodity price trends. The expansion will reach an expanded and diversified audience ranging from rural farmers, regional wholesalers, and local government units to enable them to understand better market conditions prevailing in their areas of operation. Furthermore, incorporating advanced machine learning techniques, such as LSTM networks or hybrid models that integrate ARIMA with deep learning, could further enhance the predictive abilities of the system. The methods are superior in detecting complex patterns, seasonality, and non-linear relationships within large datasets, leading to more accurate and adaptive forecasts.

Another critical factor for the success of ARIMArket is improving user accessibility and experience. The development of a mobile application with a user-friendly interface would make the system more accessible to a larger population, especially those who rely heavily on

smartphones for information. Multilingual interfaces would be used to cater to the diverse linguistic demographics of the Philippines. Personalized alerts like changes in price with major importance or forecasts based on preferences will enhance engagement and usability further, thus building better adoption of the web application.

Collaboration with government agencies, private market stakeholders, and NGOs is essential for refining the design of the system and ensuring practical application. Collaboration with regulatory bodies could help ARIMarket in monitoring price ceilings, enforcing compliance with market regulations, and supporting subsidy allocation programs. Private sector collaboration can enhance data sharing and operational insights, while NGOs can utilize the system for community-based programs focused on food security and livelihood improvement.

Finally, a structured user feedback loop would allow the continuous improvement of ARIMarket. Collecting user suggestions and monitoring the use of the system can be used to refine the model's accuracy, relevance, and functionality to meet changing needs in the market. Such regular updates, informed by user insights, would mean that ARIMarket remains a cutting-edge tool in the realm of price prediction.

5.6. Conclusion

The development of ARIMarket underscores the significant potential of predictive analytics and time series forecasting in tackling the complex challenges that commodity price fluctuations pose. Using the well-established statistical approach of the ARIMA model, the web application provides a systematic, data-driven solution for highly accurate predictions of future trends. Indeed, the whole premise of such a system starts from the generation and utilization of trustworthy historical information from the Department of Agriculture, whereby price behaviors become possible analysis with actionable input. From a strong sense of using rigorous statistical methods, the system provides clear and accurate predictability regarding essentials such as rice, meat, fish, fruits, and vegetables, considered among the bare essentials used daily by lots of household units and trade business structures.

ARIMarket is an important tool for enabling different stakeholders in different sectors. For the consumer, it allows budgeting and cost-conscious purchasing decisions that help to reduce the impact of price volatility on household finances. For businesses, the system allows efficient inventory management and pricing strategies, optimizing operations and profitability. Farmers benefit from market trend analysis, which allows them to have better harvest time planning and risk management. Policymakers also appreciate the usefulness of this web

application, since data-driven evidence is provided to inform decisions over the price controls, subsidies, and other forms of intervention intended to stabilize the market.

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